Higher Education and Corporate Innovation

Dongmin Kong, Bohui Zhang, and Jian Zhang*

Current Version: February 2022

Journal of Corporate Finance, Forthcoming

* Dongmin Kong is from School of Finance, Zhongnan University of Economics and Law; Email: kongdm@mail.hust.edu.cn. Bohui Zhang is from School of Management and Economics and Shenzhen Finance Institute, The Chinese University of Hong Kong, Shenzhen (CUHK-Shenzhen). Email: bohuizhang@cuhk.edu.cn. Jian Zhang is the corresponding author, who is from School of Business and Management, Shanghai International Studies University. Mailing address is 550 Dalian Road, Shanghai, China 200083; Email: jianzhang@shisu.edu.cn.

Higher Education and Corporate Innovation

Abstract

This paper investigates the impact of higher education on corporate innovation. To establish causality, we exploit a policy-induced exogenous shock in the supply of Chinese college-educated labor starting in 2003. Using a difference-in-differences approach, we find that Chinese firms in skilled industries generate better innovation outcomes as measured by patents and citations than those in unskilled industries. This effect is more pronounced among firms headquartered in a province with more science and engineering college graduates, young firms that are more likely to hire young graduates, and firms located near universities. Moreover, higher education expansion increases a firm's innovative human capital in terms of the number of educated employees and inventors. Finally, we show that technological innovation is a mechanism through which higher education affects productivity growth and, thus, the economy.

Keywords: corporate innovation; higher education expansion; human capital; high-skilled labor; China
JEL classification: G38; J24; O31

1 Introduction

Higher education, which occurs after the completion of secondary education, is delivered at universities and colleges as the final stage of formal learning. According to Barro and Lee (2013), only 6.7% of the population worldwide received a bachelor's degree or above by 2010. Despite its limited prevalence worldwide, higher education has been shown to play a significant role in driving economic growth.[1] However, the economic mechanisms underlying the growth effect of higher education are still unclear. For example, Aghion et al. (2009) state that "researchers most often study education and growth, neglecting intermediating variables that are likely to reveal the mechanisms at work." Given that technological innovation is vital for economic growth (Romer, 1986; Kogan et al., 2017), we attempt to fill this gap by proposing such a mechanism and investigating the impact of higher education on corporate innovation.

We posit that a larger labor force supply with higher education enhances corporate innovation. According to endogenous growth theory, human capital is the engine of the innovation process (e.g., Romer, 1990; Aghion and Howitt, 1992; Acemoglu, 1996; Aghion and Howitt, 1998). However, such a positive role of human capital in innovation may differ between higher education and primary and secondary education. The primary- and secondary-educated workforce is considered to only facilitate the imitation of existing technology, whereas college-educated labor could promote the adoption of new technology that eventually leads to more sustainable economic growth (Vandenbussche, Aghion, and Meghir, 2006; Aghion et al., 2009; Hunt and Gauthier-Loiselle, 2010; Castelló-Climent and Mukhopadhyay, 2013). For example, using Indian census data and enrollment rates at different education levels, Castelló-Climent and Mukhopadhyay (2013) show that a 1% change in higher education has the same effect on economic growth as a 13% decrease in illiteracy rates.

The effect of higher education on innovative growth is not completely unambiguous.[2] First, higher education does not always ensure a higher education quality. For example, higher education in developing countries may suffer from poor quality, which may undermine the positive effect of higher education on growth (e.g., Li, Whalley, and Xing, 2014). Second, according to USPTO (2012) and the European Commission (2013), more than 75% of patents

[1] See, e.g., Becker (1964), Nelson and Phelps (1966), Barro (2001), Goldin and Katz (2001), Gradstein and Justman (2002), Krebs (2003), Shapiro (2006), Bowlus and Robinson (2012), Gennaioli et al. (2013), and Lucas (2015).

[2] Benhabib and Spiegel (1994) and Cohen and Soto (2007) find no effect of schooling improvements on growth. Moreover, the growth effects of education tend to vary across countries. In a survey concerning education and economic growth, Krueger and Lindahl (2001) report the puzzling observation that education is positively associated with subsequent growth only in countries with the lowest level of education. Similarly, Pritchett (2001) shows that substantial investments in education yield only a small growth in payment in developing countries.

are filed by firms in traditional manufacturing industries in the US and Europe, while the inventors of these patents, i.e., blue-collar workers, mostly rely on basic education instead of higher education. Therefore, basic education might be considered a contributor to innovation in traditional industries (D'Acunto, 2014).

The above conjectures should be tested empirically to determine how higher education affects corporate innovation. The greatest challenge is the identification of a causal effect free of reverse causality and omitted variable concerns. First, the supply and demand of higher education could be endogenously determined by a region or country's economic growth and innovation. For example, stronger economic growth could lead to more investment in higher education via a relaxation of credit constraints, and alternatively, economic development could generate stronger incentives to invest in higher education in countries close to the technological frontier (Vandenbussche, Aghion, and Meghir, 2006). Second, both higher education and corporate innovation are likely to be correlated with unobserved economic, financial, and political characteristics.

To address these potential endogeneity concerns, we exploit a natural experiment, i.e., the higher education expansion policy starting in 1999 in China, that generated an exogenous surge in the supply of high-skilled labor starting in 2003 (e.g., Li, Whalley, and Xing, 2014; Knight, Deng, and Li, 2017; Che and Zhang, 2018; Xing, Yang, and Li, 2018; Ou and Hou, 2019). This setting is appealing for several reasons. First, higher education in China is under a centrally planned system in which college admission is controlled by the Ministry of Education, and education policies are decided by the central government. The motivation underlying the expansion policy in 1999 is to postpone high school graduates' entry into the labor market and stimulate domestic demand for educational services (Wan, 2006). Thus, the higher education expansion policy is largely exogenous to all firms and is not driven by firms' patenting activities.

Second, the expansion of higher education caused by this policy is sizable. For example, during the period from 1999 to 2009, the enrollment of college students in China increased by 17.2% annually from 1.55 million in 1999 to 6.49 million in 2009. This higher education expansion since 1999 resulted in a substantial increase in the supply of skilled labor force starting in 2003.

Third, the level of corporate human capital depends on both a firm's labor demand for skilled labor and the market's labor supply. The expansion policy enables us to study the impact of a supply shock in the skilled labor pool while controlling for the influence of other unobservable firm characteristics or economic conditions. Therefore, by comparing the post-shock innovation output of firms in skilled industries (i.e., treatment firms) with that of firms

in unskilled industries (i.e., control firms), we can identify the direct effect of higher education on corporate innovation.

Using a panel of 1,268 public Chinese firms from 1999 to 2009, we adopt a difference-in-differences (DID) approach to investigate how the innovation output of firms in skilled industries relative to that of firms in unskilled industries changes after the surge in the supply of college-educated workers. To alleviate the concern regarding the selection of firms in skilled industries, we classify the skilled industries in China based on the human capital intensity measure of US industries. The US leads the global economy, and the human capital intensity in the US reflects the underlying frontier technology of industries worldwide. Moreover, the US industry classification is less likely to be contaminated by Chinese policy influences, which could alleviate potential self-selection issues.

Consistent with our conjecture, we find that skilled labor has a significant positive effect on firm innovation. On average, compared with firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to a 10.6%, 8.7%, 7.7%, and 2.1% larger increase in the number of patents, number of citations, number of nonself-citations, and number of citations per patent, respectively, among firms in skilled industries. To test whether the parallel trend assumption of the DID approach holds, we follow Bertrand and Mullainathan (2003) and examine the dynamics of innovation output surrounding the introduction of the higher education expansion policy. Our test reveals that there is no pretreatment trend in innovation output. In addition, the significant impact of higher education expansion on innovation occurs two years after the policy's enactment.

Furthermore, we implement two placebo tests to investigate whether our results are purely driven by chance. First, we randomly assign firms to a pseudo treatment group and assign the remaining firms to a pseudo control group. Then, we reconduct the DID analysis based on the pseudo treatment and pseudo control groups. We repeat this procedure 5,000 times. Second, we select 2001 as the event year and reconduct our regression in the pre-2003 period. In both placebo tests, we find that the DID estimates are insignificantly different from zero.

Our findings might still be subject to the firm self-selection issue. To address this concern, we adopt a propensity score matching technique to mitigate the observable difference in the pre-shock characteristics between the treatment and control groups. We find the same quantitative results as in the full panel sample. One could argue that unobservable industry shocks may exist and relate to both human capital and corporate innovation. To mitigate this concern, we exploit the fact that most college graduates stay and work in the province where they attended college (Peking University, 2011). After controlling for industries, we designate

the treatment firms in provinces with more college graduates as the "new treatment firms" and the treatment firms in provinces with fewer graduates as the "new control firms". We obtain the same results. Our findings also hold in a battery of robustness tests performed to rule out alternative explanations using alternative subsamples, model specifications, and additional tests based on Taiwanese and US samples.

Subsequently, we identify the economic mechanism underlying our findings. These economic channels are built based on existing studies investigating how the higher education expansion could benefit the skilled labor supply of firms according to different fields of education and firm types. First, the literature shows that college graduates who majored in science and engineering (S&E) are more important for a firm's innovation than those with other majors (e.g., Hunt and Gauthier-Loiselle, 2010; Kerr and Lincoln, 2010). Consistently, we find that the positive effect of the higher education expansion on firm innovation is more significant among firms headquartered in a province with more S&E college graduates.

Second, young employees tend to join young firms with greater innovation potential and higher growth because young workers have more recent education related to current technical skills and are more risk tolerant (Ouimet and Zarutskie, 2014). We expect young firms to be more affected by the college expansion policy because young firms tend to hire more young graduates. Indeed, we find that relative to older firms, young firms in skilled industries tend to yield higher innovation output after the higher education expansion policy.

Third, firms' geographic proximity to universities may affect the effect of higher education on corporate innovation because college graduates may choose to work in local firms. For example, Zucker, Darby, and Torero (2002) find that star scientists are more likely to choose local firms than firms outside their regions. In particular, we expect firms located close to universities to be more likely to benefit from the increased supply of skilled labor. To investigate the impact of the surge in locally educated labor supply on corporate innovation, we weight the local college student admission expansion rate by the inverse distance between local universities and the firm within the same city and calculate the weighted local college admission expansion rate. The results show that higher education expansion has a more significant impact on corporate innovation in the subsample with the higher expansion rate in local universities.

To present more direct evidence that higher education expansion increases human capital stock in a firm, we examine the relationship between higher education expansion and firm employees' education levels. We show that firms in skilled industries hire more educated people (e.g., employees and inventors) than those in unskilled industries after the higher

education expansion. In addition, we investigate the fundamental nature and importance of innovation changes after the exogenous surge in the supply of skilled labor. We employ five additional measures of innovation output, namely, patent exploitation, patent exploration, patent originality, patent generality, and foreign citations. We find that firms in skilled industries have greater patent exploitation, exploration, originality, generality, and foreign citations.

Finally, we test whether technological innovation can serve as the mechanism linking higher education and economic growth. We find that the positive impact of higher education on the total factor productivity (TFP) is more pronounced in innovative firms. However, we do not find any significant impact of higher education on capital investment. This finding indicates that higher education expansion promotes productivity growth mainly by enhancing technological innovation, thus stimulating economic growth in innovative industries.

This paper makes two major contributions to the existing literature. First, our paper contributes to the literature by providing evidence that innovation serves as an important mechanism underlying the growth effect of higher education. Specifically, we identify that higher education has a strong impact on firm innovation, and through this effect, we link higher education to economic growth. Our evidence also contributes to the debates regarding the role of human capital in innovation. Although human capital is regarded as the engine of the innovation process in endogenous growth theory (e.g., Romer, 1990; Aghion and Howitt, 1992; Acemoglu, 1996; Aghion and Howitt, 1998), previous studies have reached mixed conclusions concerning the heterogeneous effects of education on innovative human capital (Aghion et al., 2009; Hunt and Gauthier-Loiselle, 2010; Castelló-Climent and Mukhopadhyay, 2013; D'Acunto, 2014).

Moreover, we contribute to the innovation literature investigating the drivers of firm innovation by identifying the role of corporate human capital. Our findings lend support to previous studies concerning the effect of managers' human capital on corporate innovation (Ederer and Manso, 2013; Balsmeier, Fleming, and Manso, 2017; Custódio, Ferreira, and Matos, 2019; Chemmanur et al., 2019). Moreover, our study adds to the growing literature concerning the determinants of corporate innovation at the employee level, including employee incentives, treatment, and efforts (Chang et al., 2015; Chen et al., 2016; Liu et al., 2017), tolerance for failure (Tian and Wang, 2014), and labor law and unionization (Acharya, Baghai and Subramanian, 2013; Flammer and Kacperczyk, 2015; Bradley, Kim and Tian, 2017; Gao and Zhang, 2017).

Our paper is related to Che and Zhang (2018), who adopt the same exogenous policy and Study the effect of higher education on total factor productivity. They argue that college-educated workers may facilitate the assimilation of the newly imported high-tech equipment, hence improving productivity. Our study examines how higher education of inventors affects corporate innovation. We directly show that higher-education expansion increases the number of inventors in the firm. Our paper identifies a channel through which higher education promotes productivity and economic growth — corporate innovation.[3]

The remainder of this paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the background, variables, and sample. Section 4 presents the empirical results. Section 5 presents the additional tests. Section 6 concludes the paper.

2 Literature review

Our paper is related to a stream of studies about the impact of universities on corporate innovation. Freitas, Marques, and Silva (2013) explores the role of university–industry collaboration for the development of innovation in Brazil. They find that the role of students and firm employees is crucial for collaboration between public research organizations and companies in emergent industries. Toivanen and Vaananen (2016) find that a positive effect of engineering education on the number of patents by studying the technical universities in Finland. Andrews (2019) find that establishing a land grant college causes an increase in local invention, including in particular agricultural inventions, in college counties relative to the runner-up counties. Hvide and Jones (2018) show that university funding rules have significant effect on patenting in Norway.

Migration policies provide an alternative way to study the impact of college graduates on innovation. Kerr and Lincoln (2010) find that higher H1-B admissions enhance innovations by inventors with Indian and Chinese names and increase immigrant science and engineering employment in H1-B dependent cities and firms relative to their peers. Hunt and Gauthier-Loiselle (2010) show that college-educated immigrants generate more innovations compared to native since they are more likely to hold science and engineering degrees. Stuen, Mobarak,

[3] Che and Zhang (2018) obtains the information of college graduates at the province level. We use micro, firm-level data to show that higher education expansion enhances corporate innovation through hiring more skilled employees and inventors. We collect college admission data at the university-major-province-year level in Shanghai, Tianjin, and Jiangsu. The data set includes the admission information of each college in a specific major in a given province during a given year, including the highest, average, and lowest scores and the admission quota of each college by major-province-year. Moreover, we collect the information of education background of each employee at the firm-year level.

and Maskus (2012) studies the contribution of doctoral students to innovation. They find that both US and international students contribute significantly to the production of knowledge at scientific laboratories. Bosetti, Cattaneo, and Verdolini (2015) show that a larger pool of migrants in the skilled professions is associated with higher levels of knowledge creation in terms of patent applications and published articles. Crown, Faggian, and Corcoran (2020) find that the share of foreign-born graduates is positively related to patents and citation-weighted patents in that region using Australian data.

The related literature studies the association between education and innovation by looking at the access of individuals to mentors or potential collaborators. Azoulary, Zivin, and Wang (2010) find that the sudden death of superstar scientists lowers the productivity of their collaborators significantly. Similarly, Jaravel, Petkova, and Bell (2018) studies the impact of team-specific capital on innovation through showing a negative effect of an inventor's premature death on their co-inventor's earnings and citation-weighted patents. Bell, Chetty, Jaravel, Petkova, and Reenen (2019) show that exposure to innovation environment has a significant impact on children to become inventors when they grow up. Our paper studies the association between higher education and corporate innovation in China.

3 Institutional background, variables, and sample

3.1 Higher education in China

Education had a long history in China since the Sui dynasty (581) lasting over 1,300 years before 1905, which is the year when the Qing dynasty decided to switch the traditional education system to a Western-style education system (Bai and Jia, 2016).[4] In China's traditional education system, the civil examination is among the most important institutions. In modern China, the education system based on college admission policies can be traced back to the 1950s. In 1950, 73 universities formed three regional alliances to centralize the admissions process, and the Ministry of Education decided to implement a centralized matching mechanism and launched the first National College Entrance Examination (also known as *Gaokao* as pronounced in Chinese) in August 1952 (Chen and Kesten, 2017).

In China, the government plays a central role in developing higher education and making education policies. The development of the education system during the post-1949 era was shaped by political leaders' efforts to establish a modern nation. Higher education development

[4] China used a civil service exam to recruit its elites, including high-level state bureaucrats and a much larger corpus of non-official gentry.

endured a highly volatile period from the 1950s to the 1970s (Huang, 2001; Hayhoe, 1995). In 1958, the central government set a goal to achieve universal access to higher education in 15 years and initiated the first large-scale expansion of higher education concurrently with the "Great Leap Forward". However, the expansion was unsustainable, and the admission rates greatly fluctuated (Wan, 2006).

The Cultural Revolution, which started in 1966 and lasted ten years, was another major event in Chinese higher education development. From 1966 to 1970, all Chinese universities were closed. In the fall of 1970, some universities enrolled new students for the first time after a four-year hiatus. Then, China adopted a new policy for selecting college students, which downgraded academic standards but emphasized political qualifications. The overall college admission rate in 1976 was even lower than that in 1957.

Higher education experienced a steady increase after the restoration of selecting students based on academic performance in 1978. In the 1990s, Chinese higher education underwent a series of reforms, including modification of the regulatory control and financing of higher education, reorganization of the higher education system through institutional mergers, restructuring of the curriculum to adapt to labor market requirements, and ending the job allocations of college graduates.

Concurrent with these reforms, the enrollment in China had experienced a rapid increase in past decades. Freeman (2010) shows that there were only 924,000 enrollments in 1993, and the number quickly increased to 5.4 million in 2006; the shares of enrollments in higher education worldwide increased from 2.7% in 1990 to 16.5% in 2006. However, the increasing intensity of the competition of the college entrance examination is consistent with mass higher education. Currently, there is a hierarchy in the Chinese higher education system with 39 top universities ("985 universities"), followed by 115 known universities ("211 universities", followed by provincial and local four-year colleges and universities, and finally tertiary vocational colleges (Jia and Li, 2020).[5] Each June, over nine million Chinese students complete *Gaokao* in their residence province with six subjects.[6] Because *Gaokao* is almost the only gateway to top universities in China, fierce competition is the natural result of the high expectations[7].

---

[5] Recently, China initiated a Double First Class University Plan, which aims to ultimately build several world class universities and disciplines by the end of 2050. In total, 137 universities were selected by the Double First Class initiative, including 42 universities that were selected in the world first-class university initiatives.

[6] The first three subjects completed by all students are Chinese, mathematics, and a foreign language. The other three subjects in the social science track include history, geography, and political science, while the three subjects in the science track include physics, chemistry, and biology.

[7] Please find a detailed discussion regarding "China's entrance exam is more competitive than ever" in

3.2 Higher education expansion since 1999

The higher education expansion since 1999 represents a typical case of government intervention in China and is consistent with the government's social and economic development policies. There are several possible reasons for the higher education expansion, but many scholars believe that the postponement of high school graduates' market entry and the stimulation of the domestic consumption of education are the most important reasons (Wei and Li, 2000).

Min Tang, an economist of the Asian Development Bank in China, wrote a proposal to Premier Rongji Zhu for the admission expansion. In his proposal, Tang estimated that doubling higher education admission over the next three years could stimulate family spending on higher education. He also suggested postponing high school graduates' entry into the labor market to alleviate the employment pressure on the labor market due to the large number of worker lay-offs in state-owned enterprises (SOEs). Tang's proposal was highly attractive to policy makers in the central government since China's economy was struggling due to the Asian Financial Crisis.

In early 1999, the Ministry of Education proposed a 20% increase in admission for that year but soon revised this magnitude to an unprecedented admission increase of 44%. In Figure 1, we plot the time trend of higher education in China. Specifically, we present the time trend of college admissions and graduates in Panel A, the time trend of the growth rate of college admissions and graduates in Panel B, and the time trend of college entrance exam attendance in Panel C.

In Panel A, we show that there is an unprecedented increase in the number of students enrolled in colleges since 1999. Since it is difficult to gain acceptance to but easy to complete Chinese colleges, most Chinese college students can graduate on time. Thus, we also observe a substantial increase in college graduates since 2003. Panel B shows that there is a spike in the college admission growth rate in 1999 and 2000, confirming the impact of higher education expansion on college admissions. Panel C shows that the college admission rate has also sharply increased since 1999. Overall, the empirical evidence indicates that there was a sudden increase in college admissions since 1999, which, in turn, led to an exogeneous surge in college graduates since 2003. Detailed information regarding the college admission rate is presented in Appendix B.

https://www.insidehighered.com/blogs/world-view/china%E2%80%99s-entrance-exam-more-competitive-ever

[Insert Figure 1 about here]

3.3 Innovation measures

Following the innovation literature, we construct two measures to gauge a firm's innovation output (Lerner, Sorensen, and Strömberg, 2011; Aghion, van Reenen, and Zingales, 2013; He and Tian, 2013; Seru, 2014; Bernstein, Giroud, and Townsend, 2016; Balsmeier, Fleming, and Manso, 2017; Howell, 2017; Moshirian et al., 2021). The first measure is the total number of patent applications filed by a firm in a given year that are eventually granted. We use the patent application year instead of the year during which the patent was granted because previous studies suggest that the application year of patents captures the actual time of innovation better than the year during which the patent was granted (Griliches, Pakes, and Hall, 1986; Hall, Jaffe, and Trajtenberg, 2001).

One concern is that a simple patent count reflects only the quantity rather than the quality of corporate innovation. In principal, an important patent is supposedly cited more frequently by other subsequent patents. Following this logic, a patent's future citations should reflect the quality of the invention (Hall, Jaffe, and Trajtenberg, 2005). This notion particularly applies to patents created in developing countries, such as China, because these countries may experience a rapid growth in patenting, while their patents are less likely to be widely cited. To assess both the patent quantity and quality, we construct the second measure by counting the number of the total citations of the patents filed by a firm in one year.

We extract the patent data from the following two sources: i) Bureau van Dijk (BvD)'s Orbis patent database, which is sourced by the Worldwide Patent Statistical Database (PATSTAT) and maintained by the European Patent Office (EPO),[8] and ii) State Intellectual Property Office of China (SIPO). PATSTAT has a much wider coverage than the National Bureau of Economic Research (NBER) Patent and Citation database, which is based solely on patent filings to the US Patent and Trademark Office (USPTO). Moshirian et al. (2021) provide a detailed comparison of PATSTAT and the NBER Patent and Citation database. The patents from PATSTAT are filed by various entities, such as public firms, private firms, governments, universities, and individuals, through 94 regional, national, and international patent offices. An important feature of PATSTAT is that the database lists most patent owners using its unique firm identifiers, and we can use these identifiers to determine the patent owners' domicile,

[8] Several studies (e.g., Chang et al., 2015; Li et al., 2016; Xie, Zhang, and Zhang, 2019; and Moshirian et al., 2020) use PATSTAT to obtain firms' patent information. One advantage of PATSTAT is that the database provides information regarding patent citations, which enables us to study the innovation quality.

industry classification, and listing status.

SIPO, which performs the same function as USPTO, collects detailed information regarding all patent applications in China (Hsu, 2009; Holmes, McGrattan, and Prescott, 2015; Fang, Lerner, and Wu, 2017). However, SIPO only provides the assignee's name without any unique identifier for the assignee. Moreover, SIPO does not identify whether a patent is filed by a listed firm's subsidiaries. Thus, if we merge the patent data from SIPO with Chinese listed firms by company name, we could underestimate the number of patents filed by Chinese listed firms. Because BvD's Orbis patent database assigns a unique identifier to assignees and links patents filed by subsidiaries to their parent firms, we are able to assign patents filed by subsidiaries to their parent firms by combining information from the SIPO and BvD's Orbis patent database.

Differing from other countries, there are three types of patents in China as follows: invention patents, utility model patents, and design patents. Invention patents are granted for new technical solutions related to products, processes, or improvements and are similar to US utility patents. Utility model patents are granted for new and practical technical solutions related to the shape or structure of a product and are similar to European and Japanese utility patents. Design patents are patents related to the appearance of a product and typically involve limited technological improvements (Tan et al., 2020; Chen et al., 2020). PATSTAT includes only invention patents and utility model patents but not design patents in China. Consistent with Tan et al. (2020), we focus only on invention patents and utility model patents in our analysis.

The patent data are subject to a truncation problem (Hall, Jaffe, and Trajtenberg, 2001; 2005). The first truncation problem arises because patent applications appear in our sample only after they are granted. However, due to the application-grant lag, many applications may not be granted if they were applied for in the last several years of the database coverage. Thus, we need to adjust the raw patent count data in our data coverage (1999-2009) using the application-grant lag distribution of 1984-1998. Specifically, we adjust the patent counts as follows:

$$NumPat_y^{adj} = \frac{NumPat_y^{raw}}{\sum_{i=0}^{Endyr-y} frac_i}$$

where y is the application year, $frac_i$ is the fraction of patents granted in i years ($1999 \leq i \leq 2009$), and *Endyr* is the last year of the data sample (e.g., 2009 in our sample). The denominator represents the fraction of (successful) patent applications in year y that are expected to have been granted by the ending year (*Endyr*) of the sample based on historical patterns.

The second type of truncation problem is related to the citation counts. Patents continue to receive citations after the end of the database coverage period. To adjust the truncation bias of the citation counts, we adopt the fixed-effect adjustment approach suggested by Hall, Jaffe, and Trajtenberg (2001) and Moshirian et al. (2021). Specifically, we divide the raw value of citation counts by the average number of citations in the same technological class during the same application year. Then, we sum the adjusted number of citations during each firm year to calculate the number of citations.

Due to the right skewness of the patent and citation counts, we calculate the natural logarithm value of one plus the patent counts (*LnPat*) to measure the innovation quantity. To consider the innovation quality, we calculate the natural logarithm value of one plus the number of citations (*LnCit*), the natural logarithm value of one plus the number of nonself-citations ($LnCit_{nonself}$), and the natural logarithm value of one plus the number of citations per patent (*LnCitPat*).

As an exogenous surge in the supply of skilled labor, higher education expansion may promote corporate innovation by increasing the human capital stock. Hombert and Matray (2018) adopt the number of inventors as a proxy for the size of human capital in a firm. Thus, to further examine the direct impact of the higher education expansion on firm innovation, we also collect inventor data from PATSTAT and SIPO. The dataset includes the names of the inventors of each patent filing eventually granted. We define *LnInventor* as the logarithm value of one plus the number of inventors filing patents in a firm in a given year. We use this variable as a proxy for the number of high skilled labors in the firm.

3.4 Human capital intensity measure

We use the human capital intensity in US industries as a proxy for the Chinese industry-level human capital intensity for several reasons. First, the US leads the global economy, and the human capital intensity in the US reflects the underlying frontier technology of industries worldwide. Second, the US industry classification is less likely to be contaminated by Chinese policy influences, which alleviates self-selection issues. Third, this strategy is similar to that adopted in previous studies, such as Hsieh and Klenow (2009) and Che and Zhang (2018). For example, Che and Zhang (2018) use the percentage of workers with a bachelor's degree or above in the US in an industry as a proxy for skilled industries in China.[9]

[9] Similarly, Hsieh and Klenow (2009) adopt the U.S. labor share as a benchmark to compare the marginal products of labor and capital across plants in China and India.

Following Belo et al. (2017) and Ghaly, Dang, and Stathopoulos (2017), we construct an index to measure the human capital intensity at the industry level using US Occupational Employment Statistics (OES) data from the Bureau of Labor Statistics (BLS) and the US Department of Labor's O*NET program classification of occupations based on the labor skill level. The O*NET program classification ranks occupations into five job zones ranging from job zone 1 (occupations that require limited or no skill) to job zone 5 (occupations that require extensive skill sets). A job zone refers to a group of occupations that are similar in education, related experience, and on-the-job training.

Specifically, we define the human capital intensity index (*HCI*) as follows:

$$\mathrm{HCI}_i = \sum_{j=1}^{O}\left(\frac{E_{ij}}{E_i} * Z_j\right),$$

where E_{ij} is the number of employees in industry i working in occupation j, E_i is the total number of employees in industry i, O is the total number of occupations in industry i, and Z_j is the O*NET program classification. We classify industries using the three-digit Standard Industrial Classification (SIC) codes for the pre-2002 sample and four-digit North American Industry Classification System (NAICS) codes for the post-2002 sample (Donangelo, 2014; Belo et al., 2017; Ghaly, Dang, and Stathopoulos, 2017).[10] Intuitively, *HCI* measures the skill level of occupations in an industry with a maximum value of five and a minimum value of one. A higher score indicates that the industry requires more human capital, while a lower score implies that the industry requires limited skill.

3.5 Control variables

We control for a vector of firm and industry characteristics that may affect a firm's innovation outcomes. Hall and Ziedonis (2001) argue that large firms may have economies of scale in generating patents. Thus, we adopt the natural log of the total assets to control for firm size (*Size*). These authors also argue that older firms and capital-intensive firms may produce more patents and citations. Therefore, we define a firm's age (*Firm age*) as the log value of the number of years since the firm was established. We use the net amount of property, plant, and equipment scaled by the total assets to control for the capital intensity (*PPE*). We include the ratio of the return on assets (*ROA*) to control for operating profitability.

We adopt R&D investment scaled by the total assets (*R&D*) to control for innovation inputs. To control for ability in financing innovation, we add cash holdings (*Cash*) and total

[10] The OES data from the BLS are classified using the three-digit SIC codes for the pre-2002 period and four-digit NAICS codes for the post-2002 period.

debt *(Leverage*), both of which are scaled by the total assets in our baseline model (Himmelberg and Petersen, 1994). To control for financial constraints, we construct Hadlock and Pierce's (2010) size-age index (*SA index)*. Particularly, the *SA index* is defined as -0.737 × *Ln(Assets)*+0.043×*Ln(Assets)*^{2}+0.04×*Firm age*. The higher the *SA index*, the more financial constraints a firm may face.

To control for industry concentration, we include the Herfindahl index based on sales (*HHI*) in our regressions. Aghion et al. (2005) find an inverted U-shaped relationship between product market competition and innovation. Thus, to control for the nonlinearity of product market competition, we add the squared Herfindahl index in our regression. To minimize the effect of outliers, we winsorize all variables at 1% and 99%. The detailed variable definitions are provided in Appendix A.

3.6 Sample and summary statistics

BvD's patent database assigns a BvD ID to the assignee of each patent if the assignee is any firm covered by the BvD database. Thus, we first collect all granted patents filed by Chinese listed firms covered by the BvD database. Then, based on the BvD ID, we merge the patent data with the Chinese firms in the BvD database. Because financial industries are highly regulated industries, we exclude financial industries from our sample.

Given that the BvD database assigns SIC and NAICS classification codes to each Chinese firm, we are able to quantify the human capital intensity of each Chinese firm using the value of its corresponding industry. We expect that firms in human-capital intensive industries are more likely affected by the exogenous surge in the supply of skilled labor. Thus, we assign a firm to the treatment group if it belongs to an industry in the top quartile of *HCI* (referred to as skilled industries) in the year prior to 2003, and we assign a firm to the control group if it is in an industry in the bottom quartile (referred to as unskilled industries) in the year prior to 2003. The event year, i.e., 2003, was the first year when students who were admitted to college through the higher education expansion in 1999 graduated and entered the labor market.

Our sample consists of 8,081 firm-year observations of 1,268 public non-financial firms from 1999 to 2009.[11] The sample period starts in 1999 because the higher education expansion began in that year. To enhance comparability between the pre-shock period and the post-shock period, our sample ends in 2009.

[11] The BvD database contains both private firms and public firms in China. However, after merging all variables, we find that public firms comprise 99% of our sample. Thus, we delete private firms from our sample. When we include these private firms in our sample, we obtain the same quantitative results.

Table 1 provides the descriptive statistics of the main variables included in our analysis.[12] On average, a Chinese firm in our sample files 1.15 patents that are successfully granted, its patents receive 1.10 citations, and each patent receives 0.11 citations. The average firm has a size of 11.9 (the logarithm value of the total assets in thousand dollars), an age of 9 years (exp(2.2)=9.02), and R&D expenditure accounting for 0.1% of the total assets. The average firm holds a sizeable amount of cash with a cash ratio of 19% of the total assets. Regarding performance, on average, a sample firm has a ROA ratio of 4.5. Regarding financial leverage, the average Chinese firm is moderately levered with a book leverage ratio of 24.8%.

Overall, the summary statistics of the key variables in our sample are comparable to those reported by Fang, Lerner, and Wu (2017). The detailed sample distribution by industry, province, and year is presented in Appendix C.

[Insert Table 1 about here]

4 Empirical results

4.1 Higher education expansion and human capital stock

One premise in our identification strategy is that the higher education expansion truly increased the human capital stock of firms. To present direct evidence supporting this assumption, we examine the relationship between the higher education expansion and firm employees' education levels. Specifically, we examine whether firms in skilled industries hire more employees with higher education levels than firms in unskilled industries in response to the higher education expansion. We conduct the DID analysis by adopting the percentage of employees with bachelor's degree or above in a firm (*Bachelor%*) as the dependent variable. We rely on a hand-collection procedure by obtaining the education information of employees from annual reports.

The regression results are presented in column 1 in Table 2. We find that firms in skilled industries experience a larger increase in employees with higher education than those in unskilled industries after the surge in skilled labor supply in the market. The result is economically significant, suggesting that compared with the firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to a 1.3% larger increase in the percentage of skilled employees among firms in skilled industries. The unconditional mean of the percentage of skilled employees in the sample is 20.3%, which is equivalent to a 6.5% (=1.3%/20.3%) increase in the percentage of skilled employees from its average level.

[12] For ease of comparison, we convert the unit of all financial variables to thousand US dollars based on the exchange rate at the fiscal year end.

We further investigate the effect of the higher education expansion on the number of inventors in the firm. Column 2 in Table 2 shows that the higher education expansion significantly increases the number of inventors by 15.03% (=$e^{0.140}$-1) in firms in skilled industries, which is again consistent with our conjecture that the higher education expansion induces an exogenous surge in the supply of skilled labor in the market.

However, if the inventors entered colleges before the higher education expansion policy in 1999, we still cannot ensure that the increase in skilled labor caused by higher education expansion contributes to corporate innovation. Given this possibility, we conjecture that inventors whose first filing patent is after 2003 are more likely to be those admitted to college under the higher education expansion. Following this logic, we construct a variable (*InventorAfter2003%*) to measure the fraction of inventors whose first filing patent is after 2003.

In column 3 in Table 2, we find that the higher education expansion promotes the hiring of inventors whose first filing patent is after 2003 by 5.5% among the firms in high skilled industries. Overall, the evidence shows that the higher education expansion truly enhances the stock of innovative human capital in firms[13].

[Insert Table 2 about here]

4.2 Univariate tests

As a preliminary analysis, we perform univariate difference-in-differences (hereafter, DID) tests. Specifically, we investigate the before-after effect of the change in innovation among firms affected by the exogenous surge in the supply of skilled labor compared with the before-after effect among firms less affected by the exogenous shock in the supply of skilled labor.

Table 3 reports the univariate DID test results. Specifically, we calculate the DID estimator by first subtracting the innovation output generated during the pre-event period from that generated during the post-event period. Columns 1 and 2 present the average differences in the treatment and control groups, respectively. Column 3 reports the DID estimator using the difference between columns 1 and 2.

The results shown in Table 3 reveal that the firms in both the treatment and control groups generate more innovation quantity and quality during the post-event period than during the pre-event period. More importantly, the DID estimates reported in column 3 are all statistically and

[13] As a robustness check, we redefine the treatment and control groups by the number of college graduates. Then we rerun the DID analysis as in Table 2. We do find a significant surge in the number of inventors, though we donot find a significant impact on the number of skilled employees. To some extent, our finding still supports the assumption that higher education expansion indeed increases the human capital of inventors in firms.

economically significant. The estimate suggests that compared with firms in unskilled industries, the exogenous shock of higher education leads to a 17.1% larger increase in the number of patents among firms in skilled industries. Similarly, compared with firms in unskilled industries, the higher education expansion results in a 15.6% larger increase in the number of citations received by firms in skilled industries. The DID results indicate that compared with the control firms, Chinese firms in skilled industries experience a substantial increase in corporate innovation due to the exogenous shock in the supply of skilled labor caused by the higher education expansion policy four years prior in 1999.

[Insert Table 3 about here]

Moreover, we compute the average changes in innovation at the firm-year level from two years before the surge in the college-educated workforce caused by the higher education expansion policy (i.e., event year -2) to one year before the shock (i.e., event year -1) and from two years before the shock to *t* years (*t*=1, 3, and 6) after the shock. Panel A in Figure 2 plots the average changes in the number of patents during event windows [-2,-1], [-2,1], [-2,3], and [-2,6].

Several findings are noteworthy. First, our sample firms experience a significant increase in the number of patents after the surge in the college-educated workforce caused by the higher education expansion. Second, the difference between the treatment and control firms from event years -2 to -1 is statistically insignificant (*p*-value=0.47). Third, after expanding the event window to [-2,1], we find that the difference between the treatment and control firms is positive and statistically significant at the 5% level (*p*-value=0.05). Fourth, further extending the event window to [-2,3] and [-2,5] yields more significant differences between the treatment and control firms (*p*-value=0.02 and *p*-value<0.01, respectively).

Panels B, C, and D in Figure 2 plot the average changes in the number of patent citations, the number of nonself-citations, and the number of citations per patent, respectively. Similar to the pattern of the number of patents, our sample firms experience a significant increase in the number of patent citations, the number of nonself-citations, and the number of citations per patent after the surge in the college-educated workforce caused by the higher education expansion policy. Similarly, we do not find a statistically significant difference in the changes in the numbers of citations, nonself-citations, and citation per patent between the treatment and control firms in event window [-2,-1]. However, as we extend the window to [-2.1], [-2.3], and [-2,6], the difference between the treatment and control firms becomes more positive and statistically significant.

[Insert Figure 2 about here]

Taken together, the univariate results suggest that the firms in skilled industries experience a more significant increase in their innovation quantity and quality than those in unskilled industries after the surge in the supply of skilled labor. This finding indicates that the surge in the supply of skilled labor has a positive effect on corporate innovation in China. To identity the specific effect of higher education on innovation, we need to perform more refined multivariate tests, which are described in the following section.

4.3 Multivariate DID analysis

We examine the effect of the higher education expansion on corporate innovation by performing a standard DID test through the following regression:

$$Innovation_{i,t+1} = \alpha_i + \beta Treat_i \times Post_t + \gamma Z_{i,t} + \delta_t + \varphi_i + \varepsilon_{i,t},$$

where *i* indexes the firm, and *t* indexes the year. $Innovation_{i,t+1}$ represents *LnPat*, *LnCit*, $LnCit_{nonself}$, and *LnCitPat* of firm *i* in year *t*+1. $Treat_i$ assumes a value of one if a firm belongs to a skilled industry or a value of zero if a firm belongs to an unskilled industry. $Post_t$ is a dummy variable that assumes a value of one for the 2003-2009 period, which is the post-period of the surge in the college-educated workforce caused by the higher education expansion policy in 1999, and zero for the 1999-2002 period. We choose 2003 as the event year because this was the first year when students who were admitted to college through the higher education expansion policy in 1999 graduated and entered the labor market.

$Z_{i,t}$ denotes a vector of control variables that may affect a firm's innovation output. In particular, we include firm size (*Size*), return on asset (*ROA*), total debt scaled by the total assets (*Leverage*), cash holding scaled by the total assets (*Cash*), asset tangibility (*PPE*), log value of the firm age (Firm age), R&D expenditures scaled by the total assets (*R&D*), financial constraint (*SA index*), the Herfindahl index (*HHI*), and the squared Herfindahl index (HHI^2) to control for the non-linear effect of product market competition on innovation outputs in our baseline regression. The detailed variable definitions are provided in Appendix A. We also include the year-fixed effect, δ_t, to control for time-specific shocks to innovation output. Moreover, we add the firm-fixed effect, φ_i, to account for time-invariant firm unobservable characteristics that may affect corporate innovation. Given that we include both year- and firm-fixed effects in our regression, the dummy variables *Post* and *Treat* are absorbed by the year- and firm-fixed effects. We adopt clustered standard errors at the firm level.

The coefficient of interest is the coefficient of the interaction term between *Treat* and *Post*, *Treat*×*Post*, which captures the innovation difference in treatment firms before and after the

surge in the supply of skilled labor rather than the similar before-after difference in the control firms. The DID estimate enables us to remove biases in the post-event comparison between the treatment and control groups. These biases could be the result of either a permanent difference between the two groups or a possible time trend effect (Imbens and Wooldridge, 2009). If we believe that the surge in the supply of skilled labor leads to a larger increase in innovation output among firms in skilled industries, this coefficient should be positive and statistically significant.

Table 4 reports the results of the multivariate DID analysis. In all specifications, the coefficient of *Treat*×*Post* is positive and statistically significant at the 1% level, indicating that compared with the firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to a larger increase in both innovation quantity and quality among the firms in skilled industries. Particularly, compared with the firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to a 10.6% (=$e^{0.101}$-1) larger increase in the number of patents among the firms in skilled industries. Furthermore, compared with the firms in unskilled industries, the exogenous shock leads to an 8.7% (=$e^{0.083}$-1) larger increase in the number of citations among the firms in skilled industries. Moreover, compared with the firms in unskilled industries, the higher education expansion results in a 7.7% (=$e^{0.074}$-1) larger increase in the number of nonself-citations and a 2.1% (=$e^{0.021}$-1) larger increase in the number of citations per patent among the firms in skilled industries. Thus, the economic significance of the impact of the higher education expansion policy on corporate innovation output is also sizable.

[Insert Table 4 about here]

Taken together, the evidence from both the univariate and multivariate analyses suggests that the exogenous surge in the supply of skilled labor has a positive effect on corporate innovation.

4.4 Identification issues

4.4.1 Trend analysis

Although the higher education expansion policy represents an exogenous shock to Chinese firms, it is still possible that our results may be driven by reverse causality due to the concern that the expected change in corporate innovation output induces the government to issue new higher-education policies. In addition to the plausible reverse causality of the shock, another assumption related to the validity of the DID estimation is the parallel trend, which requires the average change in corporate innovation to be the same in the firms in skilled

industries and those in unskilled industries in the absence of shock. Without a parallel trend, the DID estimation may generate biased results.

To address the above concerns, we follow Bertrand and Mullainathan (2003) and Atanassov (2013) and examine the dynamics of innovation output surrounding the surge in the supply of skilled labor by estimating the following equation:

$$Innovation_{i,t} = \alpha_i + \beta_1 Treat_i \times Before_{i,t}^{-3} + \beta_2 Treat_i \times Before_{i,t}^{-2} + \beta_3 Treat_i \times Before_{i,t}^{-1} + \beta_4 Treat_i \times Current_{i,t}^{0} + \beta_5 Treat_i \times After_{i,t}^{1} + \beta_6 Treat_i \times After_{i,t}^{2} + \beta_7 Treat_i \times After_{i,t}^{3+} + \gamma Z_{i,t} + \delta_t + \varphi_i + \varepsilon_{i,t},$$

where $Before_{i,t}^{-3}$ ($Before_{i,t}^{-2}$ or $Before_{i,t}^{-1}$) is a dummy variable that assumes a value of one if the observation occurs three years (two years or one year) before the surge in skilled labor supply and zero otherwise. $Current_{i,t}^{0}$ is a dummy variable that assumes a value of one if the observation occurs in the year when the surge in skilled labor supply starts and zero otherwise. $After_{i,t}^{1}$ ($After_{i,t}^{2}$ or $After_{i,t}^{3+}$) is a dummy variable that assumes a value of one if the observation occurs in the first year (the second year or the third year and beyond) after the surge in skilled labor supply and zero otherwise. All other variables are the same as those described in the baseline DID regression.

The coefficients of interest are β_1, β_2 and β_3. If there is any pre-trend in innovation between the treatment and control groups, we should obtain statistically significant β_1, β_2 and β_3 coefficients. We report the dynamic DID results in Table 5. The coefficients of β_1, β_2 and β_3 are statistically insignificant in all regressions, indicating that the parallel assumption of DID holds. The absence of any pre-trend implies that the positive effect of the increased supply of skilled labor on corporate innovation is not due to policymakers simply responding to the demand of innovative firms. To some extent, this finding mitigates the reverse causality concern.

We show that the coefficient estimates of β_4 and β_5 are small in magnitude and statistically insignificant in nearly all regressions. Moreover, the coefficients of β_6 and β_7 are larger in magnitude and statistically significant, suggesting that the impact of human capital starts to emerge two years after the sudden surge in the supply of skilled labor. This finding is consistent with the existing literature suggesting that corporate innovation is a long-term process (Hall et al., 2001; Manso, 2011; Balsmeier, Fleming, and Manso, 2017)[14].

[14] We notice that there is a significant increase in *LnCit* and *LnCitPat* in a relatively short period following the exogenous shock in the supply of skilled labor. We have three potential explanations to our findings. First, we find that the percentage of high skilled labor and inventors increase significantly following the exogenous surge in the supply of college graduate in 2003. Due to a significant increase in human capital stock, there might be a sharp

Figure 3 shows the evolution of innovation surrounding the surge in human capital supply, i.e., the difference in corporate innovation between the treatment and control groups surrounding the exogenous shock. Consistent with the dynamic DID estimations, we find that the difference between the treatment and control groups is insignificant during the three years prior to the exogenous shock, suggesting that there is no pre-trend in patents between the treatment and control groups.

Overall, the empirical evidence shows that the firms in the treatment and control groups share a similar trend in corporate innovation prior to the surge in the supply of skilled labor, thus satisfying the parallel trend assumption required by the DID estimation.

[Insert Table 5 about here]

[Insert Figure 3 about here]

4.4.2 Placebo tests

It is still possible that our DID analysis results are driven purely by chance. To address this concern, we conduct two placebo tests. First, we conduct simulations that artificially assign our sample firms to a treated group and a control group (Tan et al., 2020; Bradley, Kim, and Tian, 2017). In each simulation, we randomly draw firms from the same number of industries during the pre-event year as skilled industries, i.e., the pseudo treatment firms. Then, we treat the remaining industries as unskilled industries, i.e., the pseudo control firms.[15] Subsequently, we merge these pseudo treatment firms with the pseudo control firms to form a panel pseudo sample. Next, we conduct baseline DID regressions based on this panel pseudo sample. Finally, we repeat this simulation process 5,000 times.

In Panel A in Table 6, we summarize the distribution of the coefficients of *Treat×Post* from the DID regressions by reporting the mean, 5th percentile, 25th percentile, median, 75th percentile, 95th percentile, and standard deviation of the placebo estimates of the interaction

surge in patent quality. This is consistent with Che and Zhang (2018) that higher education expansion has an instant impact on productivity. Second, Chinese firms might be different from US firms in terms of innovation efficiency. Francis, Kim, Wang, and Zhang (2018) find that changes in job security has a significant impact on innovation one year after the passage of employment protection laws in 20 non-US OECD countries. Shao and Xiao (2019) find that corporate tax deduction has an instant impact on corporate innovation in China using a DID analysis. Li, Shan, Tian, and Hao (2020) find that corporate innovation output respond instantly to the passage of new labor law in China. Similarly, Zhang, Wu, Zhou, and Yuan (2020) find that corporate innovation output increases instantly following the construction of High-speed rail. Third, the instant impact on patent citations might be due to self-citation. For obtaining subsidies from the government, firms may have incentives to boost their innovation output in a short term. As we show in column 4 of Table 5, the patent citations increase in year t+2 following the higher education expansion, if we remove self-citation. Thus, our finding is consistent with the conjecture that higher education expansion may take a longer time on innovation output.

[15] In 2002, the pre-event year, 32 industries are classified as skilled industries, while 32 industries are classified as unskilled industries in the original sample.

term. We also report the corresponding *t*-statistics. We find that the magnitude of the placebo coefficients of the interaction term are close to zero. Moreover, the corresponding *t*-statistics are small and insignificant. This finding suggests that it is very unlikely that the DID regression results shown in Table 4 are purely driven by chance.

To further ensure that our findings are not due to chance, we conduct a second placebo test by selecting a pseudo-event year and reconduct our regression. Particularly, since 2001 is the middle year during the pre-2003 period, we select 2001 as the pseudo-event year and reconduct our regression in the pre-2003 period.[16] We find an insignificant coefficient of *Treat*×*Post*, indicating that the results of our baseline model are less likely driven by chance. The results are presented in Panel B in Table 6.

[Insert Table 6 about here]

4.4.3 Propensity score matching

Because we do not observe the counterfactual outcome of the treatment firms, we are unable to compare the outcome differences of choices in the same treatment firm to evaluate the impact of higher education expansion. Thus, we have to select a control firm identical to the treatment firm, except for the control firm is not affected by the exogenous surge of skilled labor. However, one may argue that the firms in the treatment group substantially differ from those in the control group, although this possibility is very unlikely given that the classification of skilled industries is not affected by the firms' choices. If this possibility holds, the impact of the higher education expansion shown in the DID estimation might be due to the omitted observable differences between our treatment and control groups.

To address this concern, we match each treatment firm to one control firm using propensity score matching (PSM). We first implement a logit regression to model the probability of being a treatment firm in the year prior to the shock. In the logit regression, we include a vector of control variables, including firm size, ROA, leverage, cash holdings, asset tangibility, firm age, R&D expenditures, and patent growth rate.[17] We add the patent growth rate to ensure that the parallel trend assumption of the DID regression holds (Tan et al., 2020).

We report the PSM-DID results in Table 7. Again, we find consistent results. All coefficients of the interaction term (*Treat*×*Post*) are still positive and statistically significant in

[16] The pre-2003 period is from 1999 to 2002. When we choose 2000 as the pseudo-event year, we obtain the same insignificant results. Because our dependent variable is the one-year-ahead innovation output, if we choose 2002 as the pseudo-event year, our dependent variable will be missing.

[17] The balance tests are reported in Appendix D.

the regressions. Moreover, the magnitude of the interaction term is slightly larger than that in the baseline results, suggesting that our main findings are robust to alternative samples. Particularly, compared with the firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to an 11.9% (=$e^{0.112}$-1) larger increase in the number of patents among the firms in skilled industries. Furthermore, compared with the firms in unskilled industries, the exogenous shock leads to a 9.4% (=$e^{0.09}$-1) larger increase in the number of citations among the firms in skilled industries. Moreover, compared with the firms in unskilled industries the higher education expansion results in an 8.0% (=$e^{0.077}$-1) larger increase in the number of nonself-citations and a 2.3% (=$e^{0.023}$-1) larger increase in the number of citations per patent among the firms in skilled industries.

[Insert Table 7 about here]

4.4.4 Controlling for industry shocks

Unobservable industry shocks related to both human capital and corporate innovation may exist. For example, industry escalation may drive China to move the traditional industrial sectors to high-tech industries. More firms may participate in this process, and more inventors are attracted due to the potential benefits. To mitigate this concern, we exploit the fact that most college graduates stay and work in the province where they attended college (Peking University, 2011). Thus, the surge of skilled labor in the labor market should be larger in provinces with more college graduates. However, treatment firms in provinces with either more or fewer graduates should be affected by the same unobserved industry-wide shocks. To some extent, the location of universities in China is exogenous to local economic factors. The distribution of Chinese universities is determined by the relocation of universities from the coastal to the inland areas during the 1955-1957 period for the purpose to reduce the adverse impacts of potential wars along the coast. To date, the regional distribution of universities persists.

If our documented effect is truly due to unobservable industry shocks, the change in innovation among the treatment firms in provinces with more graduates should not differ from that among treatment firms in provinces with fewer graduates. However, if our documented effect is truly due to the exogenous surge of the supply of skilled labor, the treatment firms in provinces with more graduates should be affected more than the treatment firms in provinces with fewer graduates. Because we compare firms in treatment industries in some provinces with those in the same treatment industries in other provinces, we can control for unobservable industry characteristics.

Thus, we designate the treatment firms in provinces with more college graduates "new

treatment firms" and the treatment firms in provinces with fewer graduates "new control firms". Because there might be more talented immigrants in first-tier cities, we remove the four first-tier cities, i.e., Beijing, Shanghai, Guangzhou, and Shenzhen, from our analysis. We reconduct the DID regression to compare the new treatment firms with the new control firms. Accordingly, we define *NewTreat* as a dummy variable that assumes a value of one for each treatment firm in provinces with more college graduates and zero otherwise.

Table 8 presents the results. The coefficient of the interaction term is positive and statistically significant in all regressions. We find a larger increase in firm innovation in the new treatment firms after the surge in the supply of skilled labor caused by the higher education expansion policy. This finding indicates that the observed increase in innovation following the higher education expansion is less likely to be driven by plausible industry shocks.

[Insert Table 8 about here]

4.5 Robustness checks

In this subsection, we perform a battery of tests to check the robustness of our baseline DID analysis.

First, we treat four Chinese cities, namely, Beijing, Shanghai, Guangzhou, and Shenzhen, as first-tier cities due to their superior economic performance, which enables them to attract the most talented college graduates from throughout the country. To some degree, the first-tier cities simultaneously have the most innovative firms and the most innovative human capital in China. Thus, our results might simply be due to the regional economic shocks in first-tier cities. To address this concern, we remove firms headquartered in first-tier cities and reconduct our DID analysis, and we still find a positive and highly significant coefficient of the interaction term. We report the results in Panel A in Table 9.

Second, some Chinese universities have three-year college programs. The students admitted to a three-year college program in 1999 should have graduated in 2002, which may have increased the labor supply before 2003. Although the graduates of three-year programs are much less skillful and limited, the pre-trend in innovation output may still exist. To address this concern, we remove the transition years (2002 and 2003) from our sample. The results are presented in Panel B in Table 9. Our findings remain unchanged.

Third, Che and Zhang (2018) argue that the regional distribution of industries might not be random. Regions that historically have more human capital-intensive industries may have some favorable policies to enhance the innovation output of firms. To alleviate the confounding effects caused by heterogeneous regional economic policies, we control for the province-year

fixed effect in Panel C in Table 9. Our results remain the same.

Fourth, during our sample period, China experienced several reforms of SOEs, including privatizing or merging smaller SOEs with large SOEs. Thus, our findings might be due to the potential benefits of SOE reforms. To address this concern, we include an interaction term between government ownership and the post-2003 indicator. Government ownership is measured by the total number of shares held by the government scaled by the total number of outstanding shares. The results are shown in Panel D in Table 9. The estimates of the interaction term of *Treat* and *Post* are still positive and highly significant.

Fifth, given that China entered the WTO and obtained the US Congress' grant of Permanent Normal Trade Relations status in 2001, our baseline results might be due to China's entry into the WTO. To address this concern, we conduct the same DID analysis using a Taiwanese sample as a placebo test. Because Taiwan entered the WTO in 2002, which is nearly the same time as mainland China, we should observe similar results in Taiwanese firms if WTO entry mainly drives our results. Panel E in Table 9 reports the regression results. We do not obtain any significant coefficient in the Taiwanese sample, indicating that our findings are less likely due to China's entry into the WTO.

Sixth, it is possible that our results are driven by the global trend of corporate innovation. Because the US leads the global economy, the technology trend in the US may transfer to China. Similarly, our results might be driven by the trend in countries or regions surrounding China due to common regional economic shocks. To address this concern, we reconduct the DID analysis using US firms in Panel F in Table 9. Although the coefficient of the interaction term is significant in the US sample, the sign is negative and opposite to that in the Chinese sample. Taken together, our findings are very unlikely explained by such a global trend.[18]

Seventh, like many developing economies, a large portion of economic growth and innovation takes place in private firms. If the policy shifts the employment of human capital from private sectors to public firms, focusing on public firms only might over-state the impact of higher education on corporate innovation in China as a whole. Thus, we look into the effect of higher education expansion on innovation activities in private firms in Panel G of Table 9. Specifically, we merge the patent data with Chinese private firms obtained from Chinese Industrial Enterprises Database. Because Chinese private firms' industries cannot be classified according to either SIC or NAICS, we construct the treatment and control groups based on the

[18] BvD Orbis does not report the number of nonself-citations received by a firm. Thus, we do not report the regression results of $LnCit_{nonself+1}$ in the Taiwan sample and US sample in Panels E and F in Table 8.

number of graduates at the province level. We rerun our DID analysis and find that higher education expansion still has a significant impact on corporate innovation of private firms in China.

[Insert Table 9 about here]

5 Additional tests

5.1 Economic mechanisms

In this section, we explore three plausible underlying economic channels through which the higher education expansion affects innovation output. These economic channels are built based on existing studies concerning how higher education expansion could benefit the skilled labor supply of firms according to the different fields of education and firm types. In particular, we examine three plausible economic channels by dividing our sample into two subsamples using the partitioning variable.

5.1.1 S&E graduates

The previous literature has well documented that college graduates in the S&E field contribute more to innovation; thus, we expect S&E graduates to play a significant role in our main findings. For example, Hunt and Gauthier-Loiselle (2010) find that skilled immigrants in the US create twice as many innovations as US natives, and one reason is that these US immigrants disproportionately hold S&E degrees. Kerr and Lincoln (2010) also find that higher H1-B admissions increase S&E employment and innovation due to the direct contributions of highly-skilled immigrants. Moreover, because most college graduates stay and work in the province where they attended college (Peking University, 2011), we hypothesize that the impact of the surge in skilled labor supply on corporate innovation should be more significant among firms headquartered in provinces with more college graduates in the S&E field.

We collect the number of college graduates in all fields from provincial education statistics yearbook data and calculate the percentage of college graduates in the S&E field.[19] Then, we divide our sample provinces into those with more college graduates in the S&E field and those with fewer such graduates based on the median level of college graduates in the S&E field at the province level. Finally, we merge our firm-level data of each province and reconduct the DID analysis. Because talented immigrants are more likely to flow into first-tier cities, i.e.,

[19] Since not all provincial education statistics yearbooks provide information regarding college graduates in all fields, our sample size is largely reduced.

Beijing, Shanghai, Guangzhou, and Shenzhen, in China and firms headquartered in these first-tier cities are less constraint by the supply of skilled labor from local universities, we remove the four first-tier cities from our sample in this analysis.

Table 10 reports the regression results. Interestingly, the coefficient of the interaction term *Treat×Post* is mainly significant among firms located in provinces with more S&E college graduates but not among firms located in provinces with fewer S&E college graduates. Moreover, we find that the effect of human capital on innovation in the subsample with more S&E graduates exhibits a larger economic magnitude than that in the subsample with fewer S&E graduates.

Specifically, in the subsample with more S&E graduates, compared with the firms in unskilled industries, the exogenous surge in the supply of skilled labor leads to a 9.2% (=$e^{0.088}$-1) larger increase in the number of patents among the firms in skilled industries. Furthermore, in the subsample with more S&E graduates, compared with the firms in unskilled industries, the exogenous shock leads to an 8.0% (=$e^{0.077}$-1) larger increase in the number of citations among the firms in skilled industries. Moreover, in the subsample with more S&E graduates, compared with the firms in unskilled industries, the higher education expansion results in a 6.1% (=$e^{0.059}$-1) larger increase in the number of nonself-citations and a 2.5% (=$e^{0.025}$-1) larger increase in the number of citations per patent among the firms in skilled industries.

These findings are consistent with our conjecture that the increased supply of skilled labor in the S&E field caused by the higher education expansion policy contributes more to corporate innovation than that in other fields. These results also suggest that it is skill training in the S&E field during college rather than the raw talent of college students that enhances corporate innovation.

[Insert Table 10 about here]

5.1.2 Young firms

While competing with older firms, young firms have lower chances to survive. However, conditional on surviving, Ouimet and Zarutskie (2014) find that those surviving young firms exhibit a higher average growth rate because young firms attract more young workers who possess more current technological skills due to more recent education and are more risk tolerant to take on risker projects within a firm. Assortative matching implies that young workers match young firms. Such young workers could play an important role in young firms by developing new products or establishing new methods of production. Therefore, we expect the impact of the surge in skilled labor supply to be more significant among young firms

because the college expansion shock provides more educated young labor to young firms.

To investigate the firm age mechanism through which the increased skill labor supply enhances corporate innovation, we divide our sample into two subsamples based on the sample median of firm age. Then, we reconduct the baseline regression in each subsample. Table 11 reports the regression results. The coefficient of the interaction term *Treat×Post* is only significant among young firms but not older firms. This finding indicates that young firms are more affected by the surge in skilled labor supply and helps explain why surviving young firms exhibit a higher growth rate fueled by corporate innovation.

[Insert Table 11 about here]

5.1.3 Local universities

Firms' geographic proximity to universities may affect the effect of higher education on corporate innovation because college graduates may choose to work in local firms. For example, Zucker, Darby, and Torero (2002) find that star scientists are more likely to choose local firms than firms outside their regions. Moreover, Bassanini, Brunello, and Caroli (2017) show that social pressure from local communities induces CEOs to refrain from dismissing local employees. Thus, if firms' innovation benefits from the surge of human capital, we expect firms located close to universities to be more likely to take advantage of the increased supply of skilled labor because geographic proximity enables easier matching or convenient communication between firms and graduates.

To formally validate this hypothesis, we manually obtain college admission data at the university-major-province-year level in Shanghai, Tianjin, and Jiangsu. The data include the admission information of each college in a specific major in a given province during a given year, including the highest, average, and lowest scores and the admission quota of each college by major-province-year. Because local universities assign most of their admission quota (usually 80%-90%) to the local province, we restrict our data to local universities in Shanghai, Tianjin, and Jiangsu. Then, we aggregate the admission quota of all S&E majors in each local university given the importance of college graduates majoring in S&E for firm innovation (Hunt and Gauthier-Loiselle, 2010; Kerr and Lincoln, 2010; Stuen, Mobarak, and Maskus, 2012). Thus, we are able to determine the local college student admission expansion rate, especially that of the key majors.

To investigate the impact of the surge in locally educated labor supply on corporate innovation, we weight the local college student admission expansion rate by the inverse distance between local universities and the firm within the same city and calculate the distance-

weighted local college admission expansion rate (*LocalExpansion*). We reconduct our baseline regression in the top tercile subsample and bottom tercile subsample measured by *LocalExpansion*.

The results are reported in Table 12.[20] We find a positive and significant coefficient of the interaction term in the top tercile subsample. This finding indicates that the higher education expansion policy has a stronger impact on firms located close to universities.

[Insert Table 12 about here]

5.2 Patenting characteristics

The previous results indicate that both innovation quantity and quality increase following the higher education expansion policy. We further investigate the fundamental nature and importance of the innovation changes after the exogenous surge in the supply of skilled labor. We expect that the exogenous surge in skilled-labor supply enhance the innovative ability of firms in terms of exploring more unknown knowledge. Moreover, college-educated inventors are able to read scientific articles and patent documents in foreign language, hence increasing cross-border knowledge spillover. We also tend to examine whether and how high education impacts technology breakthrough, which is proxied by the number of top cited patents. Thus, we employ five additional measures of innovation output, namely, patent exploitation, patent exploration, patent originality, patent generality, and foreign citations.

First, the following two generic types of innovation activities are widely studied in the literature: exploitative innovation and exploratory innovation (Levinthal and March, 1993; McGrath, 2001; Benner and Tushman, 2002). Exploitation refers to the extension of existing technologies and paradigms, while exploration involves experimenting with new knowledge. Specifically, we define exploitative and exploratory patents based on whether the patents adopt more existing or more new knowledge.

A patent is categorized as exploitative (exploratory) if at least 60% of its citations are based on firms' existing expertise (do not reflect a firm's existing expertise), which is defined as a combination of the firm's existing patents or the citations made by those patents within five years following Benner and Tushman (2002). Then, we calculate the exploitative intensity as the number of exploitative patents divided by the total number of patents filed in the same year (*Exploitation*). Similarly, we calculate the exploratory intensity as the number of exploratory patents divided by the total number of patents filed in the same year (*Exploration*).

[20] R&D is omitted from the regression due to a multicollinearity problem.

Second, we examine the fundamental importance and impact of the patents based on the patent originality and generality. Hall, Jaffe, and Trajtenberg (2001) propose that patents citing a wide array of technology classes are considered to have greater originality. Patents cited by a wider array of technology classes are treated as having greater generality. The originality index (*Originality*) is equal to 1-$\sum_{j}^{n_j} s_{ij}^2$, where s_{ij}^2 denotes the percentage of citations made by patent i that belongs to the patent technology class j of n_i patent classes. Thus, the originality index will have a higher value if a patent's citation belongs to many different fields. The generality index (*Generality*) is constructed similarly, but its computation relies on citations received rather than made. The generality index has a higher value if a patent is cited by subsequent patents belonging to many different technological fields.

Third, we collect the publication number of all patents citing firms' patents. Then, based on the first two digits of the publication number, we are able to identify whether the patent is granted by foreign countries. We calculate the percentage of citations received from foreign patents (*Foreign_Cite%*).

Fourth, following Lin, Liu, and Manso (2021), we construct the variable, Ln(Top X% Pat), as the natural logarithm of one plus a firm's total number of patents that are in the top X% category of the distribution of forward citations for patents granted to Chinese public firms in a given technological class and application year..

We adopt the measures of patent exploitation, exploration, originality, generality, the fraction of foreign citations, Ln(Top 5% Pat), and Ln(Top 10% Pat) as the dependent variables in the regression similar to those shown in Table 4. As reported in Table 13, the coefficients of the interaction term *Treat×Post* are positive and statistically significant for patent exploitation, exploration, originality, generality, foreign citations, Ln(Top 5% Pat), and Ln(Top 10% Pat). This finding suggests that firms in skilled industries invest more in R&D, further explore existing technologies, produce more important patents, and undertake technology breakthrough than those in unskilled industries following the exogenous shock in the supply of skilled labor in the market.

[Insert Table 13 about here]

5.3 Economic growth

Thus far, we show that college expansion stimulates corporate innovation due to the surge in the supply of skilled labor. However, the existing literature presents mixed evidence regarding the effect of education on economic growth, although some studies show that

innovation enhances economic growth by promoting productivity (Kogan et al., 2017; Chang et al., 2018). Thus, whether technological innovation can serve as an underlying mechanism that helps explain the impact of education on economic growth is unclear. If the positive effect of the higher education expansion on corporate innovation documented above leads to an improvement in productivity, we expect college expansion to result in significantly higher productivity. Importantly, we expect such an impact to be more pronounced in innovative industries.

To test this conjecture, we attempt to examine the impact of the higher education expansion on the total factor productivity (TFP). Following Levinsohn and Petrin (2003), we estimate *LnTFP* by using the residual of the following regression at the firm level:

$$y_{ijt} = \alpha_{jt} + \beta_{jt}L_{ijt} + \theta_{jt}K_{ijt} + \delta_{jt}M_{ijt} + \varepsilon_{ijt},$$

where i, j, and t represent firm, industry, and year. L is the logarithm of the number of workers, K is the logarithm of the total assets, and M is the logarithm of the expenses for material and other inputs. Our estimate of the total factor productivity captures a firm's deviation from its factor productivity. To examine the moderating effect of the innovation intensity, we perform a DID estimation by including the triple interaction term between *Treat×Post* and *LnPat* at year t in our baseline regression. In addition to TFP, we also look at a forward-looking measure of firm value (e.g., Tobin's Q). If higher education promotes firm productivity via spurring corporate innovation, such effects might be recognized by the stock market, translating into a higher Tobin's Q.

Table 14 reports the regression results. We find that the coefficient estimate of *Treat×Post* is positive and statistically significant at the 5% level in column 1, indicating that the higher education expansion results in a significant increase in a firm's TFP. We also show that the coefficient of the triple interaction term *Treat×Post×LnPat* is positive and statistically significant at the 5% level, suggesting that the effect of the higher education expansion on firms' TFP is more pronounced among firms with a higher level of innovation intensity.

However, higher education can stimulate both productivity growth and investment growth, both of which may be related to economic growth. To further investigate the impact of higher education on capital investment, we regress *Ln($K)* on *Treat×Post* and *Treat×Post×LnPat*. Interestingly, we show that the coefficient estimates of *Treat×Post* and *Treat×Post×LnPat* are both statistically insignificant in columns 3 and 4.

In column 5 and 6, the regression results indicate that the effect of the higher education expansion on firm value (Tobin's Q) is more pronounced among firms with a higher level of

innovation intensity. In contrast, higher education appears having no significant effect on Tobin's Q in firms without any innovation output.

Taken together, our findings suggest that innovation can serve as the economic mechanism by which higher education improves productivity and, in turn, economic growth.

[Insert Table 14 about here]

5.4 Entrepreneurship

Finally, entrepreneurship is also a strong engine of innovation and economic growth, making private firms particularly important. Thus, we examine whether higher education expansion influence firm entry in Table 15. Specifically, we extract firm information from Chinese Industrial Enterprises Database. We count the number of firms established at the city-year level.[21] Then, we collect the local economic factors at the city-year level. We reconstruct the treatment and control groups based on the number of colleges at the city level. Lastly, we conduct a DID analysis at the city-year level. In column 1 of Table 15, we find that higher education expansion does not induce entrepreneurs to establish more firms overall. However, since our sample includes plenty of firms with low-skilled labors, those firms may not respond to higher education expansion. Thus, we further examine whether higher education expansion particularly promote firm entries in human capital intensive industries. We further construct two variables to measure the number of firms with high-skilled labors and ones with low-skilled labors established at the city-year level, respectively. Specifically, *LnEnterNhigh* is measured by the log value of 1 plus the number of firms with at least one employee with bachelor degree or above established at the city-year level. *LnEnterNlow* is measured by the log value of 1 plus the number of firms with no employees with bachelor degree or above established at the city-year level. In column 2 and 3 of Table 15, we find that higher education expansion induces entrepreneurs to establish more firms with high-skilled labors, consistent with the conjecture that higher education expansion provides more high-skilled labor to the labor market. On the contrary, higher education expansion does not induce entrepreneurs to establish more firms with low-skilled labors.

[Insert Table 15 about here]

6. Conclusion

In this paper, we investigate the impact of human capital on firm innovation by exploiting

[21] Since we are interested the time when a firm is established, we will have a repeated cross-sectional data in a way that there is only one observation for each firm in our sample. We are not able to run a regression analysis at the firm-year level.

an exogenous shock in the surge in the supply of skilled labor in China. We find that firms in skilled industries experience a larger increase in innovation output as measured by both patent and citation counts than those in unskilled industries. We further show that such an effect is more significant among firms headquartered in a province with more S&E college graduates, young firms that are more likely to hire young graduates, and firms located close to universities. Moreover, the higher education expansion increases firms' innovative human capital in terms of the number of educated employees and inventors. Finally, the positive impact of higher education on the total factor productivity (TFP) is more pronounced in innovative industries. However, we do not find any significant impact of higher education on capital investment. Overall, our findings support our hypothesis that human capital plays an important role in stimulating economic growth through technological innovation.

More generally, our paper provides important implications for public policies aiming to enhance economic growth, which can be extended to other countries. First, Romer (2000) emphasizes that enhancing innovation requires the increasing the supply of workers with human capital. Our paper provides empirical evidence to evaluate the success of such polices in innovation through college expansion as key suppliers of workers with human capital. Second, does higher education promote innovation through selecting people originally with inventive ability or through nurturing people to have inventive ability? While existing theories predict the positive relationship between education and innovation, the causal effect of higher education on innovation is unexplored. Since both nature and nurture jointly determine the human capital of college graduates, it is hard to distinguish their impact on innovation. The association between higher education and innovation might be because college entrance exams screen talented people or because college education increases human capital stock of students. However, these two channels of positive association between higher education and innovation show different implications for policy. If higher education merely plays a role in screening talented people, the policies should aim at screening talented students. On the contrary, if higher education nurtures students with more human capital, the polices should focus on increasing the availability of higher education. Our paper shows that the impact of higher education on innovation is not due to selection but to treatment. Higher education helps transfer human capital to more students and leads to more skilled labor in the market, which promotes economic growth.

Acknowledgement

We especially thank the editor and the anonymous referee for insightful comments and suggestions. Prof. Jian Zhang acknowledges the financial support from National Natural Science Foundation of China (grant no. 71802160, 71991473) and the PRC Ministry of Education Youth Project for Humanities and Social Science Research (grant no. 18XJC630008). Prof. Dongmin Kong acknowledges the financial support from the Major Project of National Social Science Foundation of China (grant no. 21ZDA010).

References

Acemoglu, D., 1996, A microfoundation for social increasing returns in human capital accumulation, *Quarterly Journal of Economics* 111, 779-804.

Acharya, V. V., R. P. Baghai, and K. V. Subramanian, 2014, Wrongful discharge laws and innovation, *Review of Financial Studies* 27, 301-346.

Aghion, P., and P. Howitt, 1992, A model of growth through creative destruction, *Econometrica* 60, 323-351.

Aghion, P., and P. Howitt, 1998, Capital accumulation and innovation as complementary factors in long-run growth, *Journal of Economic Growth* 3, 111-130.

Aghion, P., J. Van Reenen, and L. Zingales, 2013, Innovation and institutional ownership, *American Economic Review* 103, 277-304.

Aghion, P., L. P. Boustan, C. Hoxby, and J. Vandenbussche, 2009, The causal impact of education on economic growth: Evidence from us, *Brookings papers on economic activity* 1, 1-73.

Aghion, P., N. Bloom, R. Blundell, R. Griffith, and P. Howitt, 2005, Competition and innovation: An inverted-u relationship, *Quarterly Journal of Economics* 120, 701-728.

Andrews M. Local effects of land grant colleges on agricultural innovation and output, 2019, NBER working paper.

Atanassov, J., 2013, Do hostile takeovers stifle innovation? Evidence from antitakeover legislation and corporate patenting, *Journal of Finance* 68, 1097-1131.

Azoulay, P., J. S. Graff Zivin, and J. Wang, 2010. Superstar extinction. *The Quarterly Journal of Economics* 125(2), 549-589.

Bai, Y., and R. X. Jia, 2016, Elite recruitment and political stability: The impact of the abolition of china's civil service exam, *Econometrica* 84, 677-733.

Balsmeier, B., L. Fleming, and G. Manso, 2017, Independent boards and innovation, *Journal of Financial Economics* 123, 536-557.

Barro, R. J., 2001, Human capital and growth, *American Economic Review* 91, 12-17.

Barro, R. J., and J. W. Lee, 2013, A new data set of educational attainment in the world, 1950-2010, *Journal of Development Economics* 104, 184-198.

Bassanini, A., G. Brunello, and E. Caroli, 2017, Not in my community: Social pressure and the geography of dismissals, *Journal of Labor Economics* 35, 429-483.

Becker, G. S., 1964, Human capital: A theoretical analysis with special reference to education, *National Bureau for Economic Research, Columbia University Press, New York and London*.

Bell, A., R. Chetty, X. Jaravel, N. Petkova, and J. Van Reenen, 2019, Who becomes an inventor in America? The importance of exposure to innovation. *The Quarterly Journal of Economics*, 134(2), 647-713.

Belo, F., J. Li, X. J. Lin, and X. F. Zhao, 2017, Labor-force heterogeneity and asset prices: The importance of skilled labor, *Review of Financial Studies* 30, 3669-3709.

Benhabib, J., and M. M. Spiegel, 1994, The role of human capital in economic development evidence from aggregate cross-country data, *Journal of Monetary Economics* 34, 143-173.

Benner, M. J., and M. Tushman, 2002, Process management and technological innovation: A longitudinal study of the photography and paint industries, *Administrative Science Quarterly* 47, 676-707.

Bernstein, S., X. Giroud, and R. Townsend, 2016, The impact of venture capital monitoring, *Journal of Finance* 71, 1591-1622.

Bertrand, M., and S. Mullainathan, 2003, Enjoying the quiet life? Corporate governance and managerial preferences, *Journal of Political Economy* 111, 1043-1075.

Bosetti, V., C. Cattaneo, and E. Verdolini, 2015, Migration of skilled workers and innovation: A European perspective. *Journal of International Economics* 96(2), 311-322.

Bowlus, A. J., and C. Robinson, 2012, Human capital prices, productivity, and growth, *American Economic Review* 102, 3483-3515.

Bradley, D., I. Kim, and X. Tian, 2017, Do unions affect innovation?, *Management Science* 63, 2251-2271.

Castello-Climent, A., and A. Mukhopadhyay, 2013, Mass education or a minority well educated elite in

the process of growth: The case of india, *Journal of Development Economics* 105, 303-320.
Chang, X., K. K. Fu, A. Low, and W. R. Zhang, 2015, Non-executive employee stock options and corporate innovation, *Journal of Financial Economics* 115, 168-188.
Chang, X., R. D. McLean, B. Zhang, and W. Zhang, 2018, Do patents portend productivity growth? Global evidence from private and public firms, *Working paper*.
Che, Y., and L. Zhang, 2018, Human capital, technology adoption and firm performance: Impacts of china's higher education expansion in the late 1990s, *Economic Journal* 128, 2282-2320.
Chen, C., Y. Y. Chen, P. H. Hsu, and E. J. Podolski, 2016, Be nice to your innovators: Employee treatment and corporate innovation performance, *Journal of Corporate Finance* 39, 78-98.
Chen, D., H. Gao, J. Luo, and Y. Ma, 2020, The effects of rural–urban migration on corporate innovation: Evidence from a natural experiment in china, *Financial Management* 49, 521-545.
Chen, Y., and O. Kesten, 2017, Chinese college admissions and school choice reforms: A theoretical analysis, *Journal of Political Economy* 125, 99-139.
Cohen, D., and M. Soto, 2007, Growth and human capital: Good data, good results, *Journal of Economic Growth* 12, 51-76.
Crown, D., A. Faggian, and J. Corcoran, 2020, High skilled immigration and the occupational choices of native workers: the case of Australia. *Oxford Economic Papers* 72(3), 585-605.
Custodio, C., M. A. Ferreira, and P. Matos, 2019, Do general managerial skills spur innovation?, *Management Science* 65, 459-476.
D'Acunto, F., 2014, Innovating to invest: The role of basic education, *Working paper*.
Donangelo, A., 2014, Labor mobility: Implications for asset pricing, *Journal of Finance* 69, 1321-1346.
Ederer, F., and G. Manso, 2013, Is pay for performance detrimental to innovation?, *Management Science* 59, 1496-1513.
European Commission, 2013. A Recovery on the Horizon? Annual Report on Eropean SMEs.
Fang, L. H., J. Lerner, and C. P. Wu, 2017, Intellectual property rights protection, ownership, and innovation: Evidence from china, *Review of Financial Studies* 30, 2446-2477.
Flammer, C., and A. Kacperczyk, 2016, The impact of stakeholder orientation on innovation: Evidence from a natural experiment, *Management Science* 62, 1982-2001.
Francis, B. B., I. Kim, B. Wang, and Z. Zhang, 2018, Labor law and innovation revisited. *Journal of Banking & Finance* 94, 1-15.
Freeman, Richard B., 2010, What does global expansion of higher education mean for the united states?, in Charles T. Clotfelter, ed.: *American universities in a global market* (University of Chicago Press, Chicago).
Freitas, I. M. B., R. A. Marques, and E. M. D. P. e Silva, 2013, University–industry collaboration and innovation in emergent and mature industries in new industrialized countries. *Research Policy* 42(2), 443-453.
Gao, H. S., and W. Zhang, 2017, Employment nondiscrimination acts and corporate innovation, *Management Science* 63, 2982-2999.
Gennaioli, N., R. La Porta, F. Lopez-de-Silanes, and A. Shleifer, 2013, Human capital and regional development, *Quarterly Journal of Economics* 128, 105-164.
Ghaly, M., V. A. Dang, and K. Stathopoulos, 2017, Cash holdings and labor heterogeneity: The role of skilled labor, *Review of Financial Studies* 30, 3636-3668.
Goldin, C., and L. F. Katz, 2001, The legacy of us educational leadership: Notes on distribution and economic growth in the 20th century, *American Economic Review* 91, 18-23.
Gradstein, M., and M. Justman, 2002, Education, social cohesion, and economic growth, *American Economic Review* 92, 1192-1204.
Griliches, Z., A. Pakes, and B. H. Hall, 1986, The value of patents as indicators of inventive activity, *National Bureau of Economic Research* 97-124.
Hadlock, C. J., and J. R. Pierce, 2010, New evidence on measuring financial constraints: Moving beyond the kz index, *Review of Financial Studies* 23, 1909-1940.
Hall, B. H., A. B. Jaffe, and M. Trajtenberg, 2001, The nber patent citation data file: Lessons, insights and methodological tools, *National Bureau of Economic Research*.
Hall, B. H., A. B. Jaffe, and M. Trajtenberg, 2005, Market value and patent citations, *Rand Journal of Economics* 36, 16-38.
Hall, B. H., and R. H. Ziedonis, 2001, The patent paradox revisited: An empirical study of patenting in

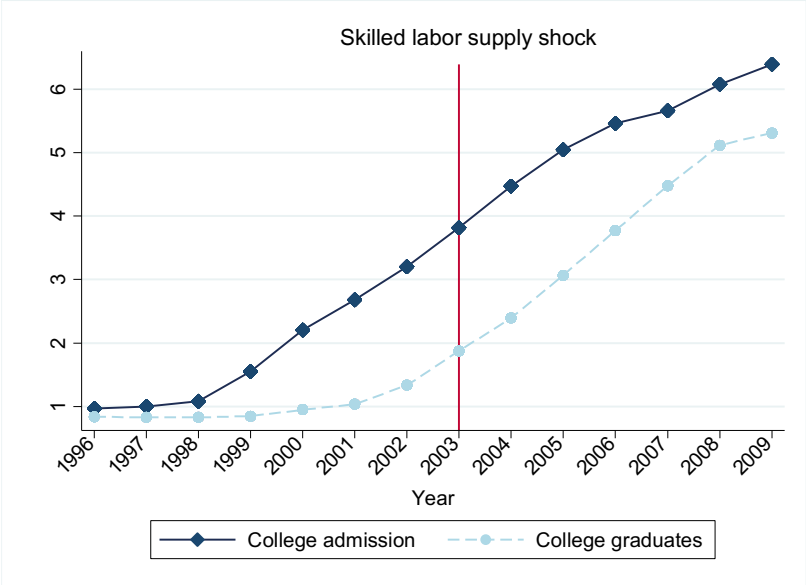
the us semiconductor industry, 1979-1995, *Rand Journal of Economics* 32, 101-128.
Hayhoe, R., 1995, An asian multiversity? Comparative reflections on the transition to mass higher education in east asia, *comparative education review* 39, 299-321.
He, J., and X. Tian, 2013, The dark side of analyst coverage: The case of innovation, *Journal of Financial Economics* 109, 856-878.
Himmelberg, C. P., and B. C. Petersen, 1994, R & d and internal finance: A panel study of small firms in high-tech industries, *Review of Economics and Statistics* 76, 38-51.
Holmes, T. J., E. R. McGrattan, and E. C. Prescott, 2015, Quid pro quo: Technology capital transfers for market access in china, *Review of Economic Studies* 82, 1154-1193.
Hombert, J., and A. Matray, 2018, Can innovation help us manufacturing firms escape import competition from china?, *Journal of Finance* 73, 2003-2039.
Howell, S. T., 2017, Financing innovation: Evidence from R&D grants, *American Economic Review* 107, 1136-64.
Hsieh, C. T., and P. J. Klenow, 2009, Misallocation and manufacturing tfp in china and india, *Quarterly Journal of Economics* 124, 1403-1448.
Hsu, P. H., 2009, Technological innovations and aggregate risk premiums, *Journal of Financial Economics* 94, 264-279.
Huang, J., 2001, The popularization of international higher education and the enrolment expansion in china's universities, *Joural of Yangzhou University (Higher Education Study Eidition)* 5, 12-17.
Hunt, J., and M. Gauthier-Loiselle, 2010, How much does immigration boost innovation?, *American Economic Journal: Macroeconomics* 2, 31-56.
Hvide, H. K., and B. F. Jones, 2018, University innovation and the professor's privilege. *American Economic Review* 108(7), 1860-98.
Imbens, G. W., and J. M. Wooldridge, 2009, Recent developments in the econometrics of program evaluation, *Journal of Economic Literature* 47, 5-86.
Jaravel, X., N. Petkova, and A. Bell, 2018, Team-specific capital and innovation. *American Economic Review* 108(4-5), 1034-73.
Jia, R. X., and H. Li, 2020, Just above the cutoff: The return to elite education in china, *Working paper*.
Kerr, W. R., and W. F. Lincoln, 2010, The supply side of innovation: H-1b visa reforms and us ethnic invention, *Journal of Labor Economics* 28, 473-508.
Knight, J., Q. Deng, and S. Li, 2017, China's expansion of higher education: The labour market consequences of a supply shock, *China Economic Review* 43, 127-141.
Kogan, L., D. Papanikolaou, A. Seru, and N. Stoffman, 2017, Technological innovation, resource allocation, and growth, *Quarterly Journal of Economics* 132, 665-712.
Krebs, T., 2003, Human capital risk and economic growth, *Quarterly Journal of Economics* 118, 709-744.
Krueger, A. B., and M. Lindahl, 2001, Education for growth: Why and for whom?, *Journal of Economic Literature* 39, 1101-1136.
Lerner, J., M. Sorensen, and P. Stromberg, 2011, Private equity and long-run investment: The case of innovation, *Journal of Finance* 66, 445-477.
Levinsohn, J., and A. Petrin, 2003, Estimating production functions using inputs to control for unobservables, *Review of Economic Studies* 70, 317-341.
Levinthal, Daniel A., and James G. March, 1993, The myopia of learning, *Strategic Management Journal* 14, 95-112.
Li, J., Y. Shan, G. Tian, and X. Hao, 2020, Labor cost, government intervention, and corporate innovation: Evidence from China. *Journal of Corporate Finance* 64, 101668.
Li, Shi, John Whalley, and Chunbing Xing, 2014, China's higher education expansion and unemployment of college graduates, *China Economic Review* 30, 567-582.
Li, X., F. Moshirian, X. Tian, and B. Zhang, 2016, The real effect of financial disclosure: International evidence. , *Working paper*.
Lin, C., S. Liu,, and G. Manso, 2021, Shareholder litigation and corporate innovation. *Management Science* 67(6), 3346-3367.
Lucas, R. E., 2015, Human capital and growth, *American Economic Review* 105, 85-88.
Manso, G., 2011, Motivating innovation, *The Journal of Finance* 66, 1823-1860.
McGrath, R. G., 2001, Exploratory learning, innovative capacity, and managerial oversight, *Academy*

of Management Journal 44, 118-131.
Moshirian, F., X. Tian, B. Zhang, and W. Zhang, 2021, Stock market liberalization and innovation, *Journal of Financial Economics* 139, 985-1014.
Nelson, R. R., and E. S. Phelps, 1966, Investment in humans, technological diffusion, and economic growth, *The American Economic Review* 56, 69-75.
Ou, D., and Y. Hou, 2019, Bigger pie, bigger slice? The impact of higher education expansion on educational opportunity in china, *Research in Higher Education* 60, 358-391.
Ouimet, P., and R. Zarutskie, 2014, Who works for startups? The relation between firm age, employee age, and growth, *Journal of Financial Economics* 112, 386-407.
Peking University, 2011, *Employment status of higher education graduates in China, 2009-2010.* (Peking University Press, Beijing).
Pritchett, Lant, 2001, Where has all the education gone?, *World Bank Economic Review* 15, 367-392.
Seru, A., 2014, Firm boundaries matter: Evidence from conglomerates and r&d activity, *Journal of Financial Economics* 111, 381-405.
Shao, Y., and C. Xiao, 2019, Corporate tax policy and heterogeneous firm innovation: Evidence from a developing country. *Journal of Comparative Economics* 47(2), 470-486.
Shapiro, J. M., 2006, Smart cities: Quality of life, productivity, and the growth effects of human capital, *Review of Economics and Statistics* 88, 324-335.
Stuen, E. T., A. M. Mobarak, and K. E. Maskus, 2012, Skilled immigration and innovation: Evidence from enrolment fluctuations in us doctoral programmes, *Economic Journal* 122, 1143-1176.
Tan, Y., X. Tian, X. Zhang, and H. Zhao, 2020, The real effect of partial privatization on corporate innovation: Evidence from china's split share structure reform, *Journal of Corporate Finance* Forthcoming, 101661.
Tian, X., and T. Y. Wang, 2014, Tolerance for failure and corporate innovation, *Review of Financial Studies* 27, 211-255.
Toivanen, O., and L. Väänänen, 2016, Education and invention. *Review of Economics and Statistics* 98(2), 382-396.
Trajtenberg, M., R. Henderson, and A. Jaffe, 1997, University versus corporate patents: A window on the basicness of invention, *Economics of Innovation and New Technology* 5, 19-50.
United States Patent Trademark Office, 2012. U.S. Patenting Trends by NAICS Industry Categories, 1963-2012.
Vandenbussche, J., P. Aghion, and C. Meghir, 2006, Growth, distance to frontier and composition of human capital, *Journal of Economic Growth* 11, 97-127.
Wan, Y., 2006, Expansion of chinese higher education since 1998: Its causes and outcomes, *Asia Pacific Education Review* 7, 19-32.
Wei, X., and W. Li, 2000, Report on the demand, supply, scale and growth speed of higher education in china., *Report of Institute of Higher Education Research, Beijing University.*
Xie, F., B. Zhang, and W. Zhang, 2021, Trust, incomplete contracting, and corporate innovation. *Management Science*, forthcoming.
Xing, C., P. Yang, and Z. Li, 2018, The medium-run effect of china's higher education expansion on the unemployment of college graduates, *China Economic Review* 51, 181-193.
Zhang, X., W. Wu, Z. Zhou, and L. Yuan, 2020, Geographic proximity, information flows and corporate innovation: Evidence from the high-speed rail construction in China. *Pacific-Basin Finance Journal* 61, 101342.
Zucker, L. G., M. R. Darby, and M. Torero, 2002, Labor mobility from academe to commerce, *Journal of Labor Economics* 20, 629-660.

Figure 1: College admissions and graduates

This figure plots the time trend of higher education in China. We collect the college admission and graduation data from Chinese Educational Statistical Yearbook and Wind. Since the higher education expansion starts in 1999, the exogeneous surge in the supply of college graduates hitting the labor market in 2003. To show the difference in college admission before and after higher education expansion, we show the time trend from 1996 till 2009

Panel A: Time trend of college admissions and graduates



Panel B: Time trend of the growth rate of college admissions and graduates

Panel C: Time trend of college entrance exam attendance

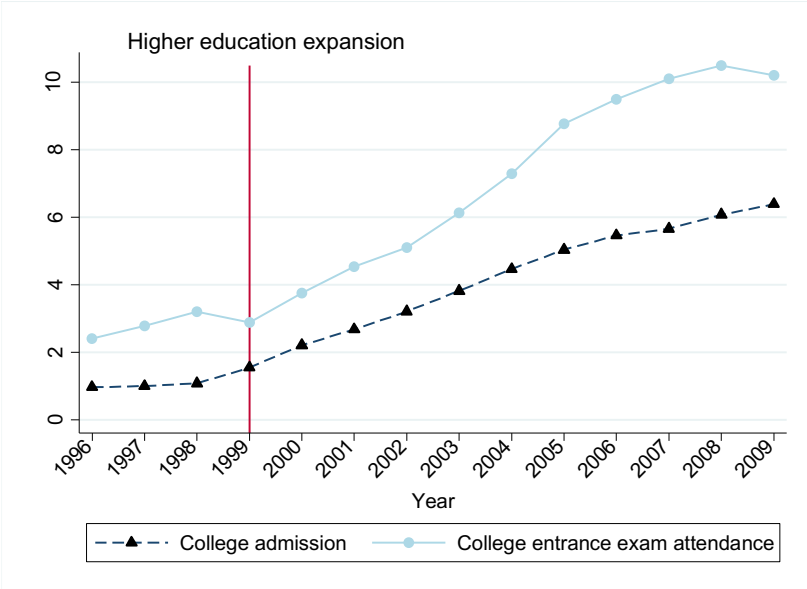
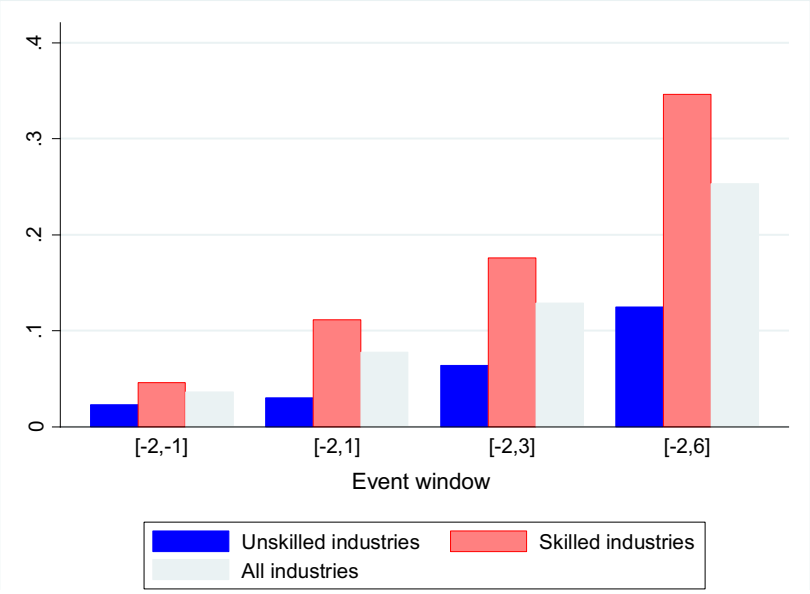


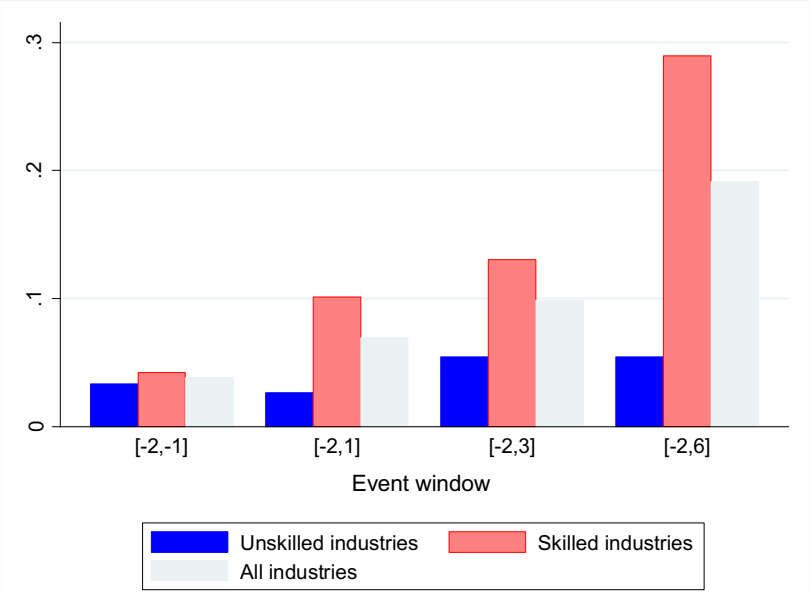
Figure 2 Average changes in innovation output around college expansion.

This figure plots the average changes in innovation output for all industries and for skilled industries and unskilled industries during 1999 to 2009. *LnPat*, *LnCit*, $LnCit_{nonself}$, and *LnCitPat* are the total number of patents, the total number of citations, the total number of nonself-citations, and the total number of citations per patent for each firm each year, respectively. The changes *in LnPat*, *LnCit*, $LnCit_{nonself}$, and *LnCitPat* are computed from two years before college expansion (i.e., year -2) to one year before college expansion (i.e., year -1) and to *t* years (*t*=1, 3, and 6) after college expansion.

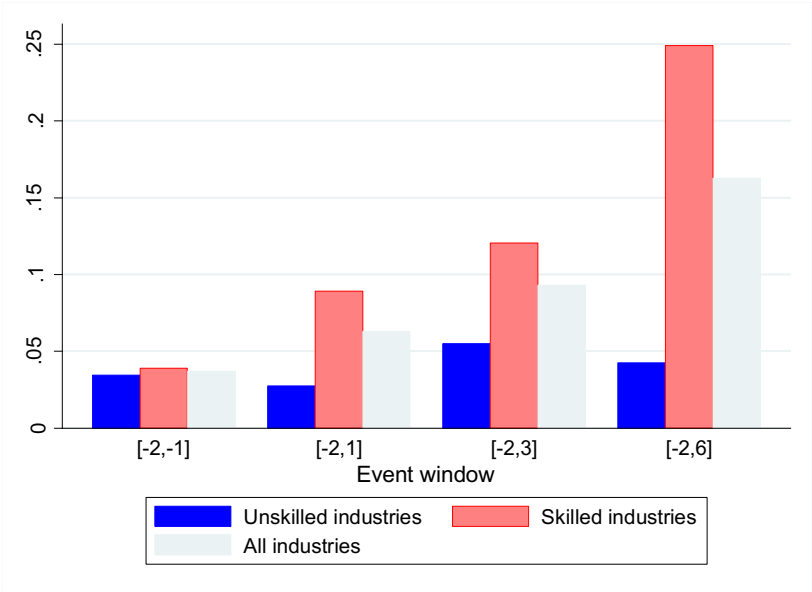
Panel A: Average changes in LnPat around college expansion



Panel B: Average changes in LnCit around college expansion



Panel C: Average changes in $LnCit_{nonself}$ around college expansion



Panel D: Average changes in LnCitPat around college expansion

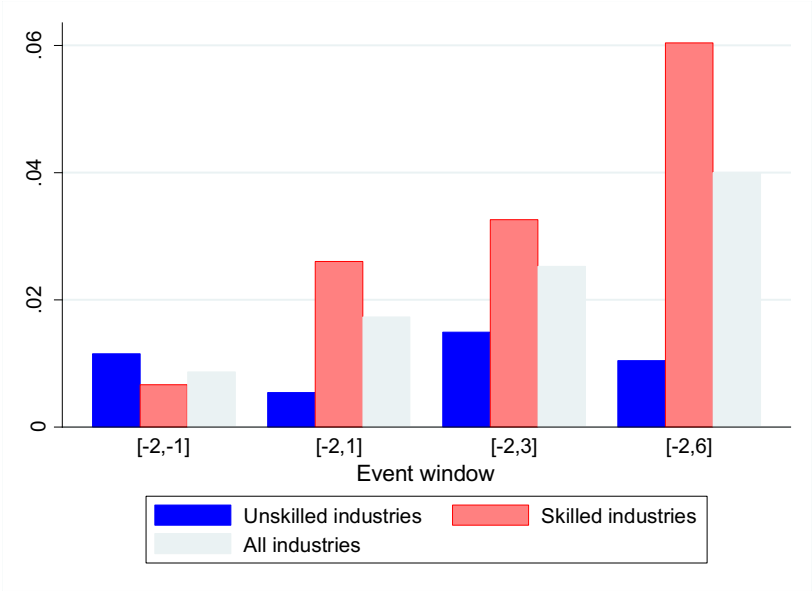
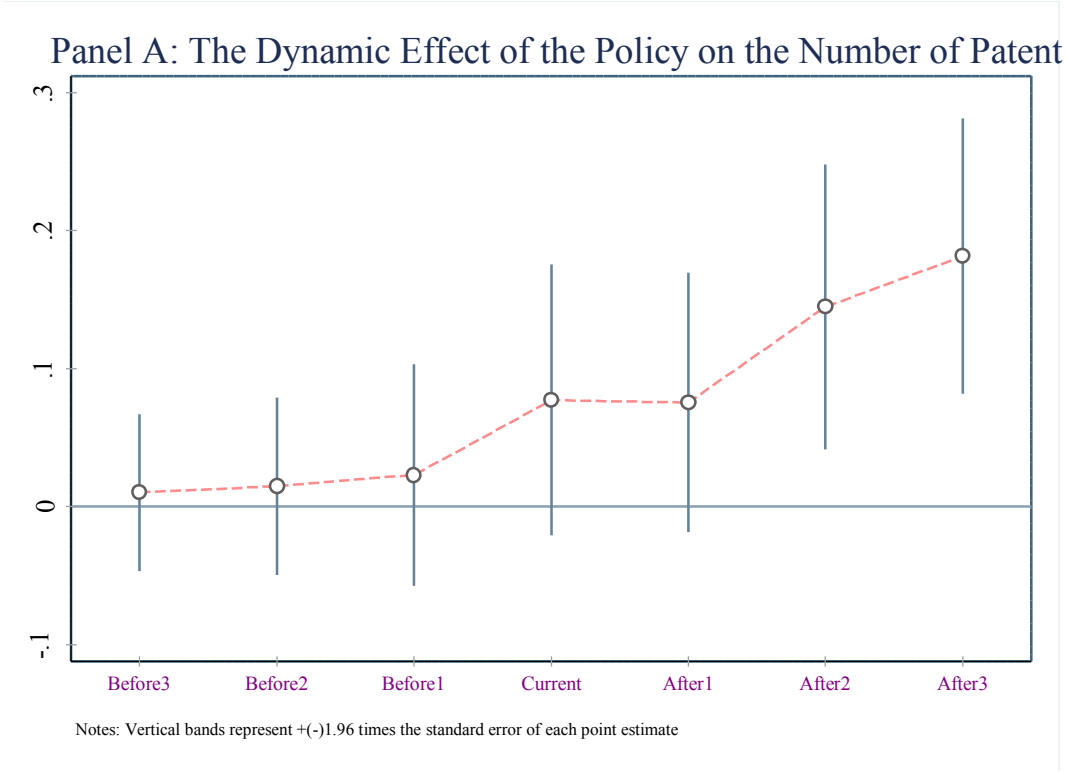
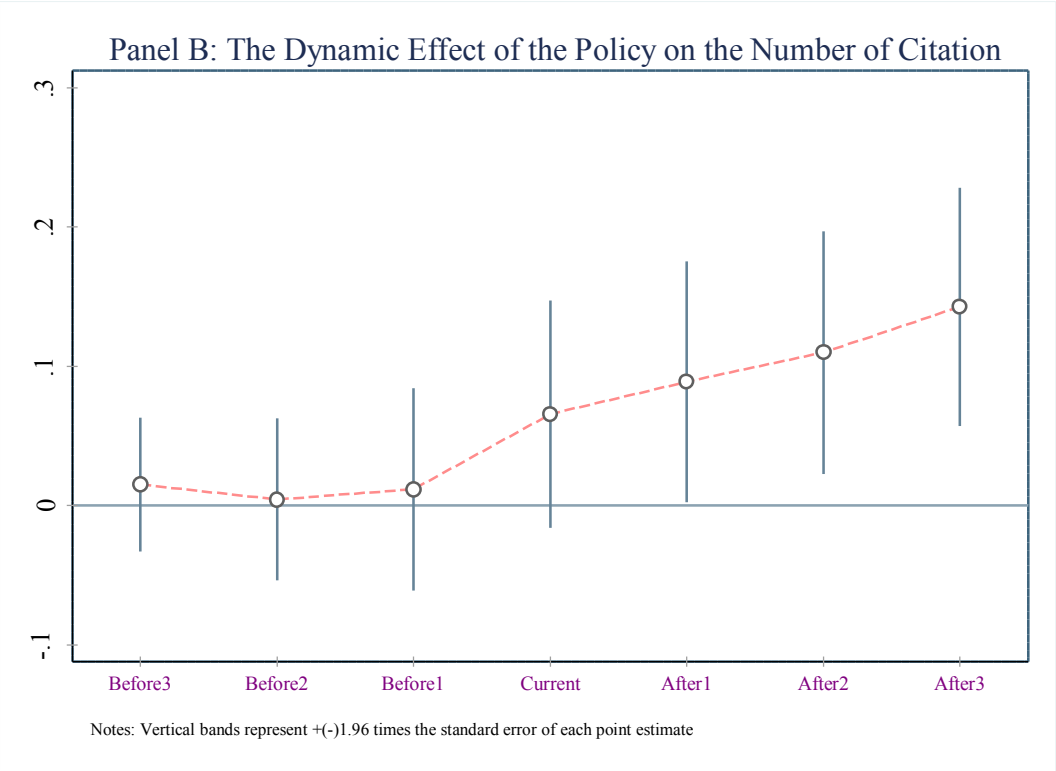
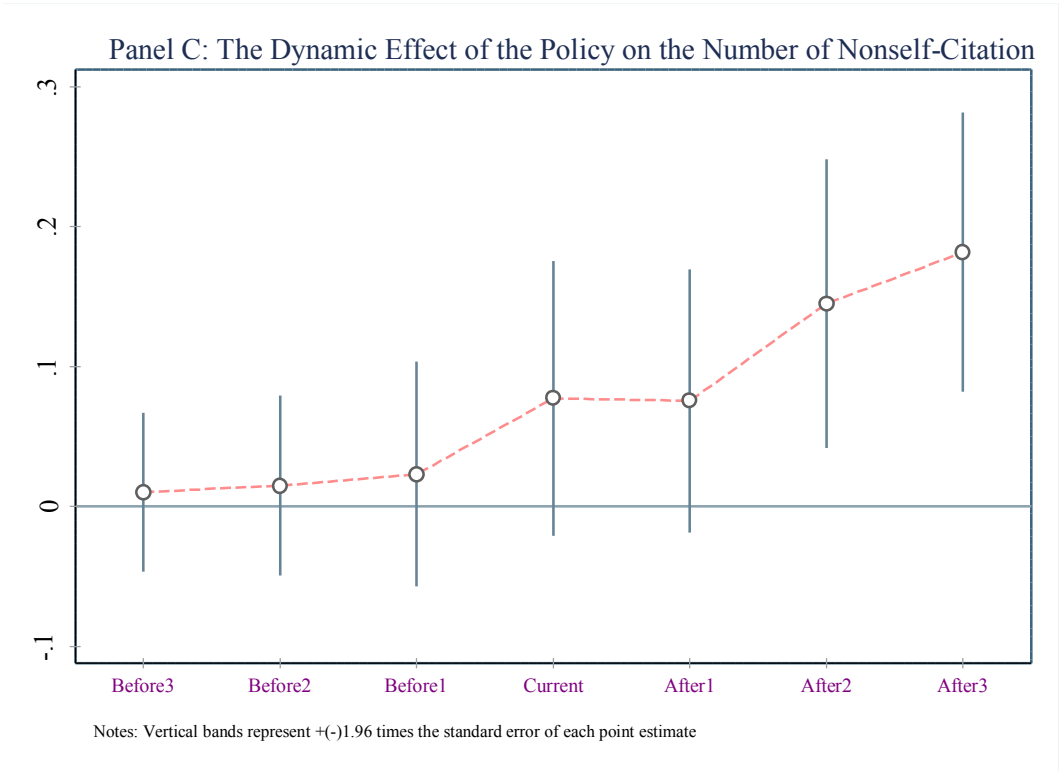


Figure 3: The dynamic effect of higher education expansion on innovation






Panel C: The Dynamic Effect of the Policy on the Number of Nonself-Citation
.3
.2
.1
0
-.1
Before3
Before2
Before1
Current
After1
After2
After3
Notes: Vertical bands represent +(-)1.96 times the standard error of each point estimate

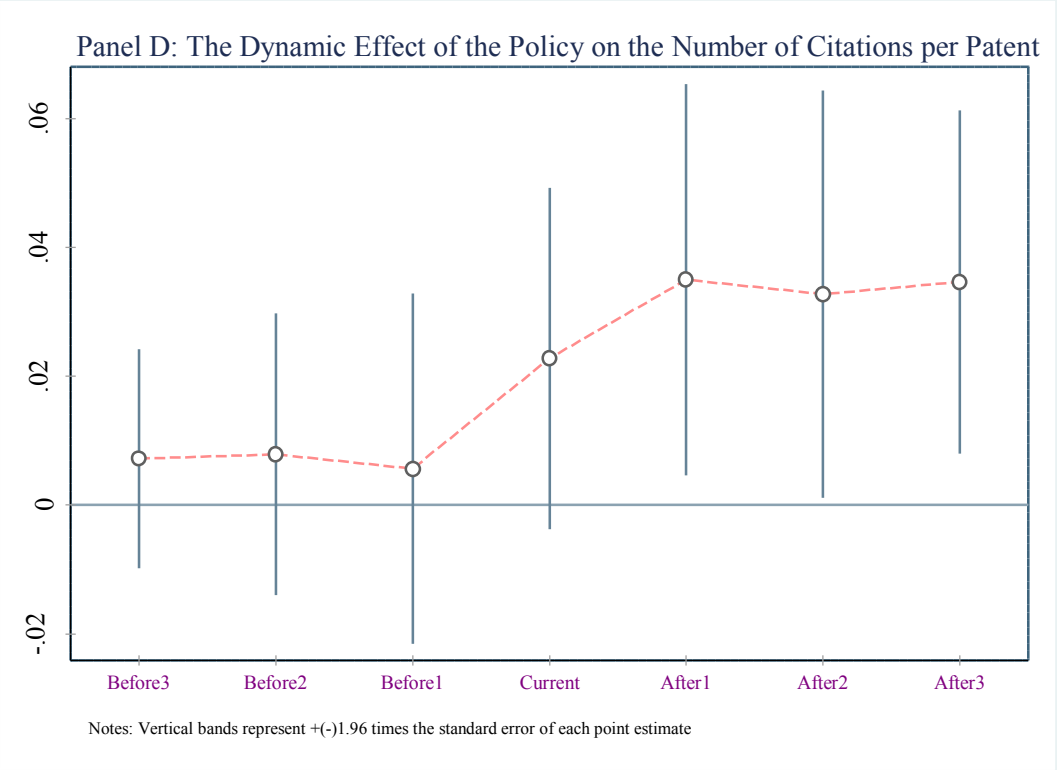

Panel D: The Dynamic Effect of the Policy on the Number of Citations per Patent
.06
.04
.02
0
-.02
Before3
Before2
Before1
Current
After1
After2
After3
Notes: Vertical bands represent +(-)1.96 times the standard error of each point estimate

Table 1 Summary Statistics

This table presents the summary statistics of the variables in our main analysis for the mean, median, standard deviation (STD), minimum (Min), 25% percentile (Q1), 75% percentile (Q3) and maximum (Max) distributions. The full panel sample comprises 8,081 firm-year observations from 1999 to 2009. Variable definitions are detailed in the Appendix A.

Variables	(1) N	(2) Mean	(3) STD	(4) Min	(5) Q1	(6) Median	(7) Q3	(8) Max
LnPat	8,081	1.148	5.376	0.000	0.000	0.000	0.000	63.713
LnCit	8,081	1.099	5.714	0.000	0.000	0.000	0.000	61.958
$LnCit_{nonself}$	8,081	0.854	4.159	0.000	0.000	0.000	0.000	37.580
LnCitPat	8,081	0.106	0.413	0.000	0.000	0.000	0.000	2.565
R&D	8,081	0.001	0.002	0.000	0.000	0.000	0.000	0.013
Size	8,081	11.938	1.268	7.160	11.185	11.899	12.699	16.290
Cash	8,081	0.191	0.141	0.004	0.09	0.156	0.253	0.728
PPE	8,081	0.308	0.183	0.003	0.171	0.285	0.418	0.788
Firm age	8,081	2.204	0.522	0.693	1.950	2.303	2.565	3.332
HHI	8,081	0.098	0.124	0.013	0.030	0.045	0.116	0.708
HHI^2	8,081	0.025	0.066	0.001	0.001	0.002	0.013	0.501
SA index	8,081	-3.012	0.406	-3.937	-3.272	-3.066	-2.824	-1.176
ROA	8,081	0.045	0.075	-0.207	0.013	0.040	0.079	0.298
Leverage	8,081	0.248	0.164	0.000	0.112	0.239	0.365	0.656

Table 2 Inventors and higher education expansion

This table reports the results of the DID analysis designed to test the impact of higher education on inventors. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)
Variables	*Bachelor%*$_{t+1}$	*LnInventor*$_{t+1}$	*LnInventorAfter2003*$_{t+1}$
Treat×Post	**0.013****	**0.140*****	**0.055*****
	(2.318)	**(3.337)**	**(3.097)**
Size	-0.007	0.079	0.054**
	(-0.588)	(1.369)	(2.268)
ROA	0.001	0.028	-0.028
	(0.075)	(0.209)	(-0.472)
Leverage	0.010	-0.047	0.001
	(0.747)	(-0.439)	(0.021)
Cash	0.025	-0.009	-0.005
	(1.532)	(-0.089)	(-0.130)
PPE	-0.012	0.183*	0.059
	(-0.778)	(1.905)	(1.604)
Firm age	-0.011	0.224***	0.094***
	(-0.904)	(2.684)	(2.693)
R&D	2.687	10.762	4.079
	(1.019)	(0.911)	(1.290)
HHI	-0.038	-1.133**	-0.426**
	(-0.737)	(-2.446)	(-2.323)
HHI^2	0.042	1.114*	0.394
	(0.565)	(1.921)	(1.638)
SA index	0.028	0.109	-0.056
	(0.806)	(0.462)	(-0.666)
Firm fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Observations	4,507	6,785	6,785
R-squared	0.951	0.767	0.689

Table 3 Univariate analysis

This table reports the results of the univariate DID analysis designed for investigating the impact of human capital on innovation. Variable definitions are reported in Appendix A. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	Treatment Diff	Control Diff	DID	Obs
	(After-Before)	(After-Before)		
$\Delta LnPat_{t+1}$	0.285***	0.114***	**0.171*** **	6,785
	(11.79)	(5.45)	**(5.32)**	
$\Delta LnCit_{t+1}$	0.234***	0.078***	**0.156*** **	6,785
	(9.87)	(4.44)	**(5.28)**	
$\Delta LnCit_{nonself\ t+1}$	0.208***	0.071***	**0.137*** **	6,785
	(9.78)	(4.25)	**(5.07)**	
$\Delta LnCitPat_{t+1}$	0.073***	0.026***	**0.047*** **	6,785
	(9.16)	(4.39)	**(4.73)**	

Table 4 Multivariate regression analysis

This table reports the results of the multivariate DID analysis designed for investigating the impact of human capital on innovation. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)		(3)
Variables	$LnPat_{t+1}$	$LnCit_{t+1}$	$LnCit_{nonself t+1}$	$LnCitPat_{t+1}$
Treat×Post	**0.101*** **	**0.083*** **	**0.074*** **	**0.021** **
	(3.401)	**(3.225)**	**(3.163)**	**(2.324)**
Size	0.065	0.044	0.055	0.018
	(1.513)	(1.171)	(1.488)	(1.136)
ROA	-0.022	0.122	0.061	0.059
	(-0.207)	(1.031)	(0.569)	(1.189)
Leverage	-0.071	0.029	0.014	0.046
	(-0.932)	(0.356)	(0.183)	(1.506)
Cash	-0.015	-0.033	-0.004	-0.005
	(-0.193)	(-0.364)	(-0.044)	(-0.153)
PPE	0.107	0.054	0.054	0.004
	(1.495)	(0.705)	(0.746)	(0.157)
Firm age	0.198***	0.129**	0.099	0.024
	(3.088)	(2.045)	(1.620)	(1.011)
R&D	7.131	10.059	9.233	2.749
	(0.790)	(1.118)	(1.199)	(1.161)
HHI	-0.765**	-0.516	-0.362	-0.138
	(-2.304)	(-1.518)	(-1.189)	(-1.210)
HHI^2	0.676	0.296	0.145	0.092
	(1.383)	(0.554)	(0.289)	(0.621)
SA index	0.090	0.041	-0.005	-0.061
	(0.536)	(0.263)	(-0.032)	(-0.949)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	6,785	6,785	6,785	6,785
R-squared	0.781	0.710	0.699	0.606

Table 5 Dynamic regression analysis

This table reports the results of the dynamic DID analysis designed for testing for pre-treatment trends and reversals. Variable definitions are reported in Appendix. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)		(3)
Variables	*LnPat*	*LnCit*	$LnCit_{nonself}$	*LnCitPat*
$Treat \times Before^{-3}$	**0.010**	**0.015**	**0.008**	**0.007**
	(0.353)	**(0.616)**	**(0.295)**	**(0.831)**
$Treat \times Before^{-2}$	**0.015**	**0.004**	**0.006**	**0.008**
	(0.453)	**(0.147)**	**(0.192)**	**(0.709)**
$Treat \times Before^{-1}$	**0.023**	**0.012**	**0.010**	**0.006**
	(0.561)	**(0.317)**	**(0.258)**	**(0.409)**
$Treat \times Current^{0}$	**0.077**	**0.066**	**0.064**	**0.023***
	(1.542)	**(1.575)**	**(1.510)**	**(1.688)**
$Treat \times After^{1}$	**0.075**	**0.089****	**0.067**	**0.035****
	(1.573)	**(2.015)**	**(1.551)**	**(2.259)**
$Treat \times After^{2}$	**0.145*****	**0.110****	**0.103****	**0.033****
	(2.751)	**(2.470)**	**(2.358)**	**(2.032)**
$Treat \times After^{3+}$	**0.182*****	**0.143*****	**0.122*****	**0.035****
	(3.569)	**(3.269)**	**(2.839)**	**(2.550)**
Size	0.081*	0.071*	0.071**	0.030**
	(1.947)	(1.847)	(1.987)	(2.243)
ROA	-0.087	-0.033	-0.021	-0.012
	(-0.775)	(-0.296)	(-0.196)	(-0.292)
Leverage	-0.088	-0.016	-0.013	0.002
	(-1.187)	(-0.230)	(-0.188)	(0.060)
Cash	0.017	0.052	0.024	0.026
	(0.244)	(0.727)	(0.357)	(0.929)
PPE	0.164**	0.114*	0.097	0.031
	(2.455)	(1.689)	(1.539)	(1.266)
Firm age	0.193***	0.127**	0.113*	0.029
	(2.990)	(2.076)	(1.894)	(1.392)
R&D	3.908	2.294	1.975	0.928
	(0.503)	(0.282)	(0.282)	(0.462)
HHI	-0.675*	-0.489	-0.362	-0.112
	(-1.832)	(-1.530)	(-1.251)	(-1.061)
HHI^2	0.660	0.419	0.287	0.136
	(1.395)	(1.060)	(0.787)	(1.230)
SA index	0.070	0.017	-0.007	-0.067
	(0.451)	(0.123)	(-0.058)	(-1.457)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	8,081	8,081	8,081	8,081
R-squared	0.742	0.666	0.650	0.574

Table 6 Placebo tests

This table reports the results of the placebo tests for DID analysis by running simulations that artificially assign our sample firms into treated group and control group. Particularly, in each simulation, we randomly draw from the pool of all firms in the pre-event year as the pseudo-treated firms. We then treat the remaining firms as the pseudo-control firms. Then we merge those pseudo-treated firms and pseudo-control firms back to the full sample to form a panel pseudo sample. Next, we conduct the baseline DID regressions based on this panel pseudo sample. Finally, we repeat this simulation process 5,000 times. We summarize the distributions of the coefficients and corresponding t-statistics for the main variable of interest, *Treat×Post*. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

Panel A: Random treatment

	Actual	Mean	P5	P25	Median	P75	P95
$LnPat_{t+1}$	**0.101*** **	-0.001	-0.068**	-0.030	-0.001	0.030	0.069**
	(3.401)	(-0.007)	(-2.196)	(-0.921)	(-0.021)	(0.927)	(2.205)
$LnCit_{t+1}$	**0.083*** **	-0.001	-0.064**	-0.027	-0.001	0.026	0.064**
	(3.225)	(-0.012)	(-2.288)	(-0.947)	(-0.010)	(0.923)	(2.257)
$LnCit_{nonselft+1}$	**0.074*** **	-0.001	-0.056**	-0.024	0.001	0.024	0.056**
	(3.163)	(-0.011)	(-2.265)	(-0.920)	(0.007)	(0.921)	(2.208)
$LnCitPat_{t+1}$	**0.021** **	-0.001	-0.021**	-0.009	0.001	0.009	0.021**
	(2.324)	(-0.012)	(-2.212)	(-0.927)	(0.010)	(0.912)	(2.156)

Panel B: Random Event

	(1)	(2)		(3)
Variables	$LnPat_{t+1}$	$LnCit_{t+1}$	$LnCit_{nonselft+1}$	$LnCitPat_{t+1}$
Treat×Post	**0.030**	**0.014**	**0.012**	**0.004**
	(0.870)	**(0.322)**	**(0.279)**	**(0.213)**
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Observations	1,243	1,243	1,243	1,243
R-square	0.798	0.697	0.696	0.633

Table 7 Propensity score matching

This table reports the results of the PSM-DID analysis. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)		(3)
Variables	$LnPat_{t+1}$	$LnCit_{t+1}$	$LnCit_{nonself t+1}$	$LnCitPat_{t+1}$
Treat×Post	**0.112*** **	**0.090*** **	**0.077*** **	**0.023*** **
	(3.581)	**(3.430)**	**(3.240)**	**(2.592)**
Size	0.030	0.037	0.049	0.011
	(0.670)	(1.130)	(1.358)	(0.936)
ROA	0.024	0.111	0.081	0.027
	(0.217)	(0.992)	(0.796)	(0.720)
Leverage	-0.088	0.002	-0.018	0.014
	(-1.130)	(0.032)	(-0.267)	(0.575)
Cash	-0.033	0.012	0.054	0.008
	(-0.422)	(0.125)	(0.594)	(0.225)
PPE	0.141*	0.074	0.067	0.022
	(1.858)	(0.911)	(0.877)	(0.872)
Firm age	0.164**	0.082	0.044	0.015
	(2.358)	(1.309)	(0.750)	(0.772)
R&D	8.423	8.211	7.329	1.011
	(0.772)	(0.760)	(0.802)	(0.444)
HHI	-0.591*	-0.481	-0.335	-0.186*
	(-1.853)	(-1.449)	(-1.136)	(-1.841)
HHI^2	0.421	0.209	0.066	0.118
	(0.873)	(0.390)	(0.131)	(0.930)
SA index	0.186	0.091	0.041	0.002
	(1.026)	(0.696)	(0.319)	(0.039)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	4,460	4,460	4,460	4,460
R-squared	0.721	0.597	0.585	0.474

Table 8 Controlling for industry shocks

This table reports the results of the multivariate DID analysis controlling for observable industry characteristics. Variable definitions are reported in Appendix. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)		(3)
Variables	$LnPat_{t+1}$	$LnCit_{t+1}$	$LnCit_{nonself t+1}$	$LnCitPat_{t+1}$
NewTreat×Post	**0.076***	**0.120*****	**0.116*****	**0.044****
	(1.825)	**(2.596)**	**(2.635)**	**(2.545)**
Size	0.091	0.079	0.131	0.050
	(0.950)	(0.700)	(1.214)	(1.010)
ROA	0.159	0.441*	0.289	0.213**
	(0.686)	(1.732)	(1.303)	(2.022)
Leverage	-0.055	0.086	0.063	0.088
	(-0.363)	(0.519)	(0.402)	(1.453)
Cash	-0.098	-0.085	-0.018	-0.014
	(-0.641)	(-0.478)	(-0.112)	(-0.232)
PPE	0.018	-0.124	-0.109	-0.040
	(0.148)	(-0.870)	(-0.797)	(-0.661)
Firm age	0.274**	0.159	0.107	0.038
	(2.410)	(1.365)	(0.931)	(0.825)
R&D	-4.001	-2.769	-0.676	-0.486
	(-0.265)	(-0.358)	(-0.079)	(-0.166)
HHI	-2.486***	-1.965**	-1.531**	-0.313
	(-2.856)	(-2.103)	(-1.989)	(-0.992)
HHI^2	2.564**	1.806	1.388	0.133
	(2.171)	(1.470)	(1.332)	(0.343)
SA index	0.000	-0.024	-0.197	-0.195
	(0.000)	(-0.052)	(-0.467)	(-0.996)
Firm fixed effect	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes
Observations	2,730	2,730	2,730	2,730
R-squared	0.777	0.716	0.700	0.594

Table 9 Robustness tests

This table reports the results of the DID analysis in alternative samples. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)		(3)
Variables	$LnPat_{t+1}$	$LnCit_{t+1}$	$LnCit_{nonselft+1}$	$LnCitPat_{t+1}$
Panel A: Excluding first-tier cities				
Treat×Post	**0.119*** **	**0.104*** **	**0.092*** **	**0.030** **
	(3.114)	**(3.048)**	**(3.010)**	**(2.436)**
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Observations	4,610	4,610	4,610	4,610
R-squared	0.782	0.708	0.694	0.582
Panel B: Deleting transition years (2002 and 2003)				
Treat×Post	**0.157*** **	**0.123*** **	**0.111*** **	**0.037** **
	(3.248)	**(2.737)**	**(2.684)**	**(2.339)**
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Observations	5,062	5,062	5,062	5,062
R-squared	0.803	0.736	0.722	0.637
Panel C: Including the province-year fixed effect				
Treat×Post	**0.105*** **	**0.084*** **	**0.071*** **	**0.023*** **
	(3.708)	**(3.440)**	**(3.305)**	**(2.651)**
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Province×Year fixed	Yes	Yes	Yes	Yes
Observations	6,634	6,634	6,634	6,634
R-squared	0.784	0.712	0.704	0.61
Panel D: Addressing the concern of SOE reforms using government ownership				
Treat×Post	**0.101*** **	**0.083*** **	**0.074*** **	**0.021** **
	(3.408)	**(3.233)**	**(3.172)**	**(2.340)**
Govown×Post	-0.022	-0.077	-0.054	-0.019
	(-0.428)	(-1.442)	(-1.233)	(-1.044)
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Observations	6,785	6,785	6,785	6,785
R-squared	0.781	0.71	0.699	0.606
Panel E: Testing the effect of WTO entry using the Taiwanese sample				
Treat×Post	**-0.08**	**-0.062**		**-0.042***

	(-0.859)	**(-0.721)**		**(-1.684)**
Controls	Yes	Yes		Yes
Firm fixed	Yes	Yes		Yes
Year fixed	Yes	Yes		Yes
Observations	2,155	2,155		2,155
R-squared	0.783	0.623		0.432
Panel F: Testing the global trend using the US sample				
Treat×Post	**-0.129*** **	**-0.189*** **		**-0.081*** **
	(-3.043)	**(-3.705)**		**(-3.630)**
Controls	Yes	Yes		Yes
Firm fixed	Yes	Yes		Yes
Year fixed	Yes	Yes		Yes
Observations	14,629	14,629		14,629
R-squared	0.882	0.849		0.586
Panel G: Higher education and corporate innovation in private firms				
Treat×Post	**0.048***	**0.052****	**0.048****	**0.024****
	(1.818)	**(2.347)**	**(2.418)**	**(2.465)**
Controls	Yes	Yes	Yes	Yes
Firm fixed	Yes	Yes	Yes	Yes
Year fixed	Yes	Yes	Yes	Yes
Observations	461,523	461,523	461,523	461,523
R-squared	0.550	0.500	0.488	0.425

Table 10 S&E graduates

This table reports the results of the DID analysis designed to test the impact of college graduates in S&E field on innovation. Variable definitions are reported in Appendix. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)		(2)				(3)	
	$LnPat_{t+1}$		$LnCit_{t+1}$		$LnCit_{nonselft+1}$		$LnCitPat_{t+1}$	
Variables	Less S&E	More S&E	Less S&E	More S&E	Less S&E	More S&E	Less S&E	More S&E
Treat×Post	**-0.007**	**0.088****	**-0.008**	**0.077****	**-0.020**	**0.059***	**-0.010**	**0.025***
	(-0.065)	**(1.992)**	**(-0.064)**	**(2.110)**	**(-0.198)**	**(1.862)**	**(-0.211)**	**(1.847)**
Size	0.276**	-0.053	0.228	-0.025	0.309	-0.026	0.094	-0.002
	(2.095)	(-0.772)	(1.281)	(-0.306)	(1.509)	(-0.346)	(1.284)	(-0.068)
ROA	0.202	0.310	0.094	0.303	0.085	0.093	0.113	0.171
	(0.391)	(1.510)	(0.160)	(1.270)	(0.157)	(0.522)	(0.505)	(1.426)
Leverage	-0.058	-0.104	-0.161	-0.031	-0.097	-0.026	-0.005	0.040
	(-0.233)	(-0.932)	(-0.614)	(-0.261)	(-0.398)	(-0.243)	(-0.048)	(0.924)
Cash	-0.112	-0.063	-0.388	-0.002	-0.097	-0.029	-0.129	0.020
	(-0.422)	(-0.394)	(-1.571)	(-0.008)	(-0.419)	(-0.156)	(-1.506)	(0.285)
PPE	0.103	0.024	0.056	-0.022	0.043	-0.032	0.037	-0.006
	(0.743)	(0.208)	(0.412)	(-0.164)	(0.307)	(-0.250)	(0.579)	(-0.102)
Firm age	-0.062	0.285**	-0.126	0.245*	-0.312	0.221*	-0.085	0.081**
	(-0.377)	(2.400)	(-0.595)	(1.914)	(-0.967)	(1.896)	(-0.848)	(2.455)
R&D	2.467	-2.162	2.287	-1.627	1.663	-2.435	-0.115	1.216
	(0.533)	(-0.311)	(0.490)	(-0.298)	(0.418)	(-0.590)	(-0.063)	(0.742)
HHI	0.509	0.054	0.461	0.170	0.865	0.257	0.191	-0.072
	(0.753)	(0.071)	(0.654)	(0.194)	(0.993)	(0.306)	(0.664)	(-0.244)
HHI^2	-0.817	-0.800	-0.814	-1.101	-1.263	-1.199	-0.294	-0.118
	(-0.870)	(-0.600)	(-0.838)	(-0.702)	(-1.025)	(-0.775)	(-0.738)	(-0.329)
SA index	-0.512	0.212	-0.432	0.141	-0.666	0.148	-0.258	-0.031
	(-1.521)	(0.895)	(-0.919)	(0.508)	(-1.260)	(0.591)	(-1.499)	(-0.322)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	861	1,770	861	1,770	861	1,770	861	1,770
R-squared	0.842	0.829	0.796	0.751	0.789	0.743	0.666	0.630

Table 11 Young firms

This table reports the results of the DID analysis designed to test the impact of young firms on innovation. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)		(2)				(3)	
	$LnPat_{t+1}$		$LnCit_{t+1}$		$LnCit_{nonselft+1}$		$LnCitPat_{t+1}$	
Variables	Young firms	Old firms	Young firms	Old firms	Young firms	Old firms	Young firms	Old firms
Treat×Post	**0.125*** **	**0.107**	**0.074** **	**0.115**	**0.069** **	**0.101**	**0.026* **	**0.027**
	(2.824)	**(1.540)**	**(2.189)**	**(1.446)**	**(2.195)**	**(1.433)**	**(1.843)**	**(1.069)**
Size	0.076	0.027	0.021	0.034	0.030	0.065	0.005	0.014
	(1.100)	(0.456)	(0.311)	(0.496)	(0.465)	(0.896)	(0.146)	(0.523)
ROA	0.098	-0.119	0.247	-0.078	0.209	-0.126	0.087	-0.008
	(0.610)	(-0.682)	(1.538)	(-0.369)	(1.362)	(-0.622)	(1.279)	(-0.112)
Leverage	-0.143	0.088	0.005	0.134	0.011	0.105	0.087*	0.041
	(-1.285)	(0.658)	(0.042)	(0.905)	(0.097)	(0.743)	(1.744)	(0.746)
Cash	-0.055	0.043	-0.046	0.037	-0.028	0.116	0.017	0.012
	(-0.426)	(0.329)	(-0.325)	(0.221)	(-0.208)	(0.816)	(0.332)	(0.216)
PPE	0.140	0.202	0.091	0.160	0.102	0.142	0.019	0.024
	(1.247)	(1.309)	(0.755)	(0.905)	(0.881)	(0.867)	(0.358)	(0.470)
Firm age	0.058	0.609**	0.108	0.422	0.092	0.335	0.015	0.083
	(0.537)	(2.515)	(1.025)	(1.431)	(0.919)	(1.077)	(0.383)	(0.605)
R&D	0.445	3.009	4.631	2.265	3.998	3.004	1.429	0.314
	(0.067)	(0.425)	(1.035)	(0.486)	(1.045)	(0.568)	(0.848)	(0.163)
HHI	-1.033**	-0.489	-0.636	-0.234	-0.512	-0.255	-0.222	0.015
	(-2.356)	(-0.694)	(-1.309)	(-0.348)	(-1.110)	(-0.418)	(-1.398)	(0.055)
HHI^2	0.799	0.699	0.277	0.415	0.158	0.460	0.163	0.109
	(1.069)	(0.769)	(0.327)	(0.465)	(0.193)	(0.565)	(0.853)	(0.272)
SA index	0.035	0.219	0.111	0.069	0.046	0.004	-0.058	-0.032
	(0.138)	(0.946)	(0.369)	(0.261)	(0.165)	(0.015)	(-0.418)	(-0.324)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	3,772	3,013	3,772	3,013	3,772	3,013	3,772	3,013
R-squared	0.792	0.843	0.741	0.764	0.728	0.750	0.631	0.675

Table 12 Local universities

This table reports the results of the DID analysis designed to test the impact of local universities on innovation. Variable definitions are reported in Appendix. All regressions include firm and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)		(2)				(3)	
	$LnPat_{t+1}$		$LnCit_{t+1}$		$LnCit_{nonselft+1}$		$LnCitPat_{t+1}$	
Variables	Low expansion	High expansion	Low expansion	High expansion	Low expansion	High expansion	Low expansion	High expansion
Treat×Post	**-0.034**	**0.515****	**-0.034**	**0.171***	**-0.034**	**0.154***	**-0.013**	**0.072***
	(-0.140)	**(2.208)**	**(-0.262)**	**(1.990)**	**(-0.262)**	**(1.735)**	**(-0.262)**	**(1.807)**
Size	-0.311	1.244	-0.480	0.746	-0.480	0.578	-0.184	0.127
	(-0.588)	(1.388)	(-0.418)	(1.051)	(-0.418)	(0.923)	(-0.418)	(0.511)
ROA	0.543	-0.408	0.638	-0.119	0.638	-1.173	0.244	0.416
	(0.512)	(-0.334)	(0.440)	(-0.102)	(0.440)	(-1.225)	(0.440)	(0.573)
Leverage	-0.583	-0.153	0.111	-0.129	0.111	-0.309	0.042	0.152
	(-0.330)	(-0.252)	(0.169)	(-0.306)	(0.169)	(-0.783)	(0.169)	(0.739)
Cash	-0.375	-0.038	1.416	0.231	1.416	-0.032	0.542	0.338
	(-0.251)	(-0.057)	(0.456)	(0.405)	(0.456)	(-0.068)	(0.456)	(1.397)
PPE	0.239	-0.690	2.648	-0.385	2.648	-0.548	1.013	-0.172
	(0.122)	(-1.076)	(0.440)	(-0.825)	(0.440)	(-1.364)	(0.440)	(-0.561)
Firm age	0.481	-0.104	0.720	0.031	0.720	0.007	0.275	0.111
	(0.516)	(-0.263)	(0.462)	(0.112)	(0.462)	(0.030)	(0.462)	(1.067)
HHI	-1.668	-3.799	2.613	-4.381*	2.613	-3.098	0.999	-3.411***
	(-0.300)	(-1.551)	(0.373)	(-1.790)	(0.373)	(-1.570)	(0.373)	(-2.959)
HHI^2	6.196	3.353	-6.747	3.941	-6.747	2.769	-2.580	3.244***
	(0.320)	(1.238)	(-0.346)	(1.562)	(-0.346)	(1.333)	(-0.346)	(2.812)
SA index	1.243	-3.489	1.696	-2.100	1.696	-1.600	0.649	-0.431
	(0.489)	(-1.343)	(0.387)	(-1.035)	(0.387)	(-0.894)	(0.387)	(-0.572)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	117	195	117	195	117	195	117	195
R-squared	0.962	0.822	0.934	0.869	0.932	0.884	0.895	0.815

Table 13 Innovation strategy and higher education expansion

This table reports the results of the multivariate DID analysis designed for investigating the impact of human capital on innovation strategy. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Variables	*Exploitation$_{t+1}$*	*Exploration$_{t+1}$*	*Originality$_{t+1}$*	*Generality$_{t+1}$*	*Foreign_Cite%$_{t+1}$*	*Ln(Top 5% Pat)t+1*	*Ln(Top 10% Pat)t+1*
Treat×Post	**0.005*** **	**0.026*** **	**0.012*** **	**0.010** **	**0.013*** **	**0.011*** **	**0.010** **
	(3.079)	**(2.932)**	**(2.946)**	**(2.293)**	**(2.648)**	**(2.581)**	**(2.447)**
Size	0.001	0.020*	0.005	0.011	0.008	-0.007	-0.007
	(0.289)	(1.710)	(0.796)	(1.498)	(1.020)	(-1.114)	(-1.037)
ROA	-0.009	0.004	-0.008	0.005	-0.019	0.053**	0.053**
	(-1.141)	(0.085)	(-0.453)	(0.218)	(-0.895)	(2.261)	(2.218)
Leverage	0.002	0.020	0.000	0.008	-0.025	0.020*	0.020*
	(0.479)	(0.801)	(0.030)	(0.478)	(-1.507)	(1.679)	(1.701)
Cash	-0.000	0.027	0.007	-0.005	0.033**	-0.020	-0.017
	(-0.004)	(1.229)	(0.514)	(-0.284)	(1.993)	(-1.304)	(-1.061)
PPE	-0.002	0.043**	0.002	0.001	0.035***	0.006	0.006
	(-0.682)	(2.236)	(0.172)	(0.067)	(3.050)	(0.463)	(0.509)
Firm age	0.001	0.046**	0.025***	0.018	0.034***	-0.014	-0.014
	(0.366)	(2.487)	(2.622)	(1.575)	(3.091)	(-1.381)	(-1.400)
R&D	0.763	-0.339	1.736	1.061	0.002	1.369	1.373
	(1.427)	(-0.309)	(1.092)	(0.951)	(0.003)	(1.141)	(1.141)
HHI	-0.007	-0.127	-0.086*	-0.006	-0.020	-0.100*	-0.097*
	(-0.602)	(-1.325)	(-1.757)	(-0.088)	(-0.310)	(-1.937)	(-1.876)
HHI^2	0.000	0.135	0.091*	-0.038	0.015	0.106*	0.103*
	(0.021)	(1.184)	(1.701)	(-0.427)	(0.190)	(1.946)	(1.884)
SA index	-0.002	-0.018	0.013	-0.023	0.007	0.018	0.014
	(-0.228)	(-0.448)	(0.658)	(-0.868)	(0.273)	(0.823)	(0.635)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	6,785	6,785	6,785	6,785	6,785	6,653	6,653
R-squared	0.399	0.544	0.378	0.552	0.540	0.388	0.393

Table 14 Higher education, Innovation, and economic growth

This table reports the results of the moderating effect of innovation on higher education and economic growth. Variable definitions are reported in Appendix A. All regressions include firm and year fixed effects. The t-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	Ln(TFP)		*Ln($K)*		*TobinQ*	
Variables	(1)	(2)	(3)	(4)	(5)	(6)
Treat×Post	**0.104****	0.094*	**-0.011**	-0.013	**-0.057**	-0.077
	(2.237)	(1.962)	**(-0.330)**	(-0.382)	**(-1.126)**	(-1.511)
LnPat		0.133*		-0.032		-0.066
		(1.668)		(-0.746)		(-0.829)
Treat×LnPat		-0.137		0.014		-0.026
		(-1.588)		(0.294)		(-0.271)
Post×LnPat		-0.143*		0.045		-0.042
		(-1.945)		(1.271)		(-0.639)
Treat×Post×LnPat		**0.159***		**-0.016**		**0.144***
		(1.953)		**(-0.400)**		**(1.670)**
Size	0.631***	0.634***	0.816***	0.817***	-0.954*	-0.963*
	(8.118)	(8.244)	(12.160)	(12.127)	(-1.824)	(-1.835)
ROA	2.065***	2.059***	-0.239	-0.238	2.122***	2.108***
	(9.225)	(9.182)	(-1.516)	(-1.504)	(4.214)	(4.180)
Leverage	0.137	0.123	-0.028	-0.027	-0.046	-0.048
	(1.053)	(0.939)	(-0.338)	(-0.321)	(-0.264)	(-0.274)
Cash	0.292**	0.290**	-0.029	-0.025	-0.059	-0.057
	(2.328)	(2.329)	(-0.307)	(-0.266)	(-0.285)	(-0.272)
Tangible	-0.692***	-0.694***	3.773***	3.772***	0.115	0.122
	(-5.296)	(-5.352)	(29.184)	(29.146)	(0.535)	(0.564)
Firm age	0.148	0.139	0.137**	0.133**	0.213	0.222*
	(1.627)	(1.551)	(2.027)	(1.978)	(1.594)	(1.674)
R&D	5.188	4.742	-3.597	-3.606	16.905	16.891
	(0.892)	(0.804)	(-1.107)	(-1.132)	(0.895)	(0.892)
HHI	-0.634	-0.313	-0.111	-0.098	0.620	0.626
	(-1.269)	(-0.659)	(-0.260)	(-0.228)	(0.963)	(0.968)
HHISQ	0.659	0.360	-0.103	-0.117	-0.713	-0.712
	(1.079)	(0.600)	(-0.211)	(-0.238)	(-0.913)	(-0.909)
SA	-0.351*	-0.352*	0.359*	0.353*	1.259	1.281
	(-1.682)	(-1.712)	(1.868)	(1.832)	(0.975)	(0.987)
Firm fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Observations	5,742	5,742	5,742	5,742	5,398	5,398
R-squared	0.891	0.891	0.962	0.962	0.590	0.590

Table 15 Higher education and entrepreneurship

This table reports the results of the DID analysis designed to test the impact of higher education on entrepreneurship. *LnEnterN* is measured by the log value of 1 plus the number of firms established at the city-year level. $LnEnterN_{high}$ is measured by the log value of 1 plus the number of firms with at least one employee with bachelor degree or above established at the city-year level. $LnEnterN_{low}$ is measured by the log value of 1 plus the number of firms with no employees with bachelor degree or above established at the city-year level. *LnGDP* is the log value of GDP per capita at the city-year level. *LnPopulation* is the log value of the number of residential with Hukou at the city-year level. *LnWage* is the log value of average monthly salary at the city-year level. All regressions include city and year fixed effects. The *t*-statistics reported in parentheses are based on standard errors clustered at the firm level. ***, **, and * indicate significance at the 1%, 5% and 10% levels, respectively.

	(1)	(2)	(3)
VARIABLES	*LnEnterN*	$LnEnterN_{high}$	$LnEnterN_{low}$
NewTreat×Post	0.127	0.127***	0.025
	(0.869)	(3.039)	(0.182)
LnGDP	-0.038	0.192**	-0.187
	(-0.156)	(2.486)	(-0.734)
LnPopulation	-0.384	-0.058	-0.327
	(-1.629)	(-0.610)	(-1.228)
Lnwage	0.069*	-0.047**	0.105***
	(1.690)	(-2.149)	(2.745)
City fixed effect	Yes	Yes	Yes
Year fixed effect	Yes	Yes	Yes
Observations	823	823	823
R-squared	0.719	0.906	0.773

Appendix A: Variable Definition

Variables	Definition	Data Source
LnPat	Log value of one plus the number of patents	PATSTAT and SIPO
LnCit	Log value of one plus the number of citations	PATSTAT and SIPO
$LnCit_{nonself}$	Log value of one plus the number of nonself-citations	PATSTAT and SIPO
LnCitPat	Log value of one plus the number of citations per patent	PATSTAT and SIPO
Exploitation	The percentage of exploitative patent in the total number of patents filed in a year	PATSTAT and SIPO
Exploration	The percentage of exploratory patent in the total number of patents filed in a year	PATSTAT and SIPO
Originality	The median of originality scores of all patents filed by firm i in year t. Following Trajtenberg et al. (1997), we define the originality score of a patent as one minus the Herfindahl index of the technology category distribution of all patents that have been cited by this particular patent.	PATSTAT and SIPO
Generality	The median of generality scores of all patents filed by firm i in year t. Following Trajtenberg et al. (1997), we define the generality score of a patent as one minus the Herfindahl index of the technology category distribution of all patents that have cited this particular patent.	PATSTAT and SIPO
LnInventor	Log value of one plus the number of inventors filing the patent application	PATSTAT and SIPO
Treat	Dummy variable taking the value of one if a firm belongs to the top quartile of industry skill measured by LSI in 2002, and taking the value of zero if a firm belongs to the bottom quartile of industry skill measured LSI in 2002.	BLS and O*Net Program
Post	Dummy variable taking the value of one if year is after 2003, zero otherwise	
Size	Log value of total asset; Total asset is measured in million dollars	BvD Osiris
ROA	Return on Asset	BvD Osiris
Leverage	Total debt divided by total asset	BvD Osiris
Cash	Cash holding divided by total asset	BvD Osiris
PPE	PPE expense divided by total asset	BvD Osiris
Firm age	Log value of firm age	BvD Osiris
R&D	R&D expense divided by total asset	BvD Osiris
HHI	Herfindahl index based on sales	BvD Osiris
HHISQ	Square term of HHI	BvD Osiris
SA index	SA index computed as in Hadlock and Pierce (2010)	BvD Osiris
Bachelor%	The percentage of employees with bachelor degree or above	RESSET

Appendix B: College admission

Year	Participants (million)	Admission (million)	Admission rate
1996	2.41	0.97	0.402
1997	2.78	1.00	0.360
1998	3.20	1.08	0.338
1999	2.88	1.60	0.556
2000	3.75	2.21	0.589
2001	4.54	2.68	0.590
2002	5.10	3.20	0.627
2003	6.13	3.82	0.623
2004	7.29	4.47	0.613
2005	8.77	5.04	0.575
2006	9.50	5.46	0.575
2007	10.1	5.66	0.560
2008	10.5	5.99	0.570
2009	10.2	6.29	0.617

Appendix C Sample distribution
Panel C1: By industry (classified by two-digits SIC) and year

	Year											
SIC	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	Total
07	3	3	3	3	4	4	4	5	7	7	8	51
10	4	6	6	8	11	16	17	19	19	19	19	144
13	4	4	4	4	4	4	3	3	3	3	3	39
15	7	7	7	9	9	10	10	10	12	13	13	107
16	2	3	5	5	5	5	5	7	9	12	13	71
20	2	2	3	4	4	4	4	4	6	6	6	45
22	16	20	24	26	27	33	33	33	35	35	36	318
23	13	18	25	27	29	33	33	34	42	46	48	348
24	0	0	1	1	1	1	1	1	1	1	1	9
25	1	1	2	2	2	3	3	2	3	4	5	28
27	3	3	3	4	3	4	5	6	6	8	8	53
28	68	94	107	129	138	143	147	171	208	233	239	1,677
29	1	2	2	3	4	4	4	4	6	6	6	42
31	0	0	1	1	1	1	1	1	1	2	2	11
34	3	4	4	5	6	6	6	7	12	16	17	86
35	31	48	62	68	73	78	81	99	136	160	170	1,006
36	34	45	58	65	71	86	88	98	123	144	159	971
37	8	13	17	21	22	25	25	29	30	33	33	256
38	5	6	7	8	10	14	14	23	38	44	53	222
41	1	2	2	2	2	2	2	2	2	3	3	23
44	7	11	13	13	14	15	17	17	17	19	19	162
45	4	5	6	7	7	8	8	8	8	7	7	75
46	1	1	2	2	2	3	3	3	3	3	3	26
47	13	19	25	30	32	33	35	35	37	38	38	335
48	4	6	7	9	9	8	8	8	10	12	14	95
50	14	18	20	22	22	23	24	27	32	35	36	273
51	11	12	15	14	14	13	12	15	16	16	18	156
53	27	30	33	38	37	37	37	40	40	42	41	402
54	1	1	1	2	2	2	2	3	6	6	7	33
58	1	2	4	4	4	5	5	6	6	6	6	49
70	10	12	13	16	17	17	17	18	18	19	19	176
73	13	18	27	32	39	42	52	68	92	116	138	637
75	1	1	3	3	3	3	3	3	3	3	3	29
79	2	2	4	4	4	4	4	3	3	3	3	36
87	0	0	1	2	3	5	5	10	17	22	25	90
Total	315	419	517	593	635	694	718	822	1,007	1,142	1,219	8,081

Panel C2: By province and year

	Year											
Province	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	Total
Anhui	5	11	14	18	19	21	23	27	29	32	33	232
Beijing	34	47	63	73	79	84	91	109	141	176	207	1,104
Chongqing	2	3	7	8	9	8	9	9	13	14	14	96
Fujian	9	10	13	16	16	20	20	26	37	41	45	253
Gansu	2	3	3	3	4	5	5	5	6	7	7	50
Guangdong	36	45	56	63	67	77	75	93	128	148	158	946
Guangxi	6	9	10	10	10	10	10	13	13	12	12	115
Guizhou	2	2	3	4	4	4	4	4	6	6	6	45
Hainan	5	6	8	9	9	9	8	9	11	12	12	98
Hebei	5	6	7	9	10	10	9	10	16	17	18	117
Heilongjiang	4	5	5	6	6	6	5	5	6	8	9	65
Henan	5	5	7	11	11	13	16	17	18	22	23	148
Hubei	12	17	19	21	20	24	24	26	33	37	38	271
Hunan	10	14	15	17	20	23	25	26	29	32	34	245
Jiangsu	17	23	34	45	47	55	57	65	84	100	105	632
Jiangxi	4	6	7	8	8	7	8	9	9	10	11	87
Jilin	12	15	18	17	18	18	18	18	20	21	21	196
Liaoning	19	21	23	23	23	23	22	26	30	33	35	278
Neimenggu	4	5	5	5	5	5	5	5	5	6	8	58
Ningxia	4	4	4	4	4	4	4	4	4	4	4	44
Qinghai	0	1	3	3	3	4	4	3	3	3	3	30
Shaanxi	13	14	15	16	15	18	18	20	20	21	23	193
Shandong	15	22	27	30	33	35	37	47	59	64	66	435
Shanghai	36	51	63	71	75	80	83	92	100	107	109	867
Shanxi	2	3	3	4	5	4	3	3	4	6	6	43
Sichuan	9	14	17	19	20	22	26	29	34	35	37	262
Tianjin	10	13	16	17	18	20	21	23	26	27	27	218
Xinjiang	6	7	7	9	9	9	9	8	9	9	10	92
Xizang	4	4	4	4	4	4	4	5	5	6	6	50
Yunnan	2	6	7	8	10	11	11	14	15	15	14	113
Zhejiang	20	26	33	41	52	59	62	70	91	107	114	675
Total	314	418	516	592	633	692	716	820	1,004	1,138	1,215	8,058

Appendix D: Balance tests of PSM

	Treatment	Control	Diff.	T-test	P-value
Size	11.97	11.95	0.02	0.29	0.774
ROA	0.017	0.019	-0.002	-0.43	0.664
Leverage	0.277	0.279	-0.002	-0.21	0.836
Cash	0.189	0.188	0.001	0.08	0.934
PPE	0.336	0.351	-0.015	-0.98	0.329
Firm age	2.058	2.018	0.040	1.02	0.307
R&D	0.0003	0.0004	-0.0001	-0.66	0.513
Patent Growth	0.005	0.028	-0.023	-0.71	0.478